SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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Assignment IV-A

CO4: Assess network and web-application security, including web vulnerabilities and wireless authentication, and recommend protective measures using data-driven analysis of authorized assessments.

Evaluation note: This Pair-work assignment: each question has shared work plus identified Student A and Student B responsibilities. Award marks for a working, keyed data model, correct R operations with before/after evidence, reconciliation/validation checks, and defended joint answers — and verify that BOTH students' individual components are present. All scanning assumes an isolated lab (e.g. 192.168.56.0/24). Accept equivalent correct R.

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Department : CSE

Sub Code / Name : ITA1405 / ETHICAL HACKING

Date of Submission : 11/11/2026

Total Marks:

M SANA FATHIMA

Student Signature **Faculty Signature**

ETHICAL HACKING

ASSIGNMENT 4 – CO4 – PART I

NETWORK, WEB AND WIRELESS SECURITY ASSESSMENT ANALYSIS

Course: ITA14 – Ethical Hacking

Course Outcome – CO4: Assess network and web-application security, including web vulnerabilities and wireless authentication, and recommend protective measures using data-driven analysis of authorized assessments.

Nature of Assignment: Individual **Duration:** 1 Week **Total Marks:** 50
Submission Date: _____11/11/2026_____

PURPOSE OF THE ASSIGNMENT

This assignment evaluates your ability to analyse the data produced by an authorized network, web-application and wireless security assessment before drawing conclusions and recommending protective measures. You must explore the assessment data, describe and visualise it, form hypotheses, test them and defend evidence-based security decisions.

You are expected to demonstrate the following progression:

> **Understand the Data → Describe → Visualise → Hypothesise → Test → Interpret → Recommend**

All analysis must be performed in R. Base R functions and standard packages may be used. All assessment data must come from authorized or lab environments only, and no exploitation is performed.

This assignment does not assess whether you can run t.test() or aov(); it assesses whether you can choose, apply and defend the right analysis to support a security decision.

TASK 1 – What Is the "Typical" Value? Central Tendency Under Skewed Security Data — 6 Marks

Web-Application Response Time

A security team runs an authorized load and response-time assessment of a web application and records the HTTP response time (in milliseconds) of every request

during the test window. Most requests complete in 200–400 ms, but a small number of slow requests take much longer.

The product/marketing team wants to advertise:

> "Average response time: 320 ms."

The security/operations team argues that this figure is misleading.

Task 1.1 – Predict and Diagnose | 2 Marks

Before performing any calculations, predict the relative positions of the mean, median and mode for the response-time data.

Your response must:

- Predict the likely ordering of the mean, median and mode for a positively skewed response-time distribution.
- Explain how skewness and extreme (slow) response times influence these measures.
- Identify at least two limitations of using a single measure to represent the responsetime experience.
- Justify why the prediction is a hypothesis that must be verified using the assessment data.

Task 1.2 – Design the Datasets | 2 Marks

Construct three response-time datasets using R.

Your design must include:

- Dataset 1 – Positively Skewed: Generate at least 500 response-time records with a realistic right tail of slow requests.
- Dataset 2 – Approximately Symmetric: Generate response times without a heavy tail.
- Dataset 3 – Edge Case: Design a dataset that exposes a limitation of a single centraltendency measure (for example, a bimodal distribution produced by two server nodes).

Document the dataset sizes, generation methods and expected statistical behaviour.

Task 1.3 – Implement and Compare | 1 Marks

Write R code to calculate the following measures for all three datasets:

- Mean
- Median
- Mode using a self-defined R function
- 10% trimmed mean
- 20% trimmed mean

Generate a histogram and a boxplot for each dataset, and compare the calculated measures.

Explain how skewness, repeated values and extreme (slow) requests affect each measure.

Task 1.4 – Design Decision and Defence | 1 Marks

The team intends to publish a single "typical response time" on its status page.

Recommend one statistical measure for each dataset and identify the most defensible single figure.

Your defence must:

- Compare the robustness of the mean, median, mode, and trimmed mean.
- Use numerical results and graphical evidence to justify the recommendation.
- Explain the trade-off between representing the usual user experience and the worstcase slow requests.
- State any limitations of the selected measure.

Evidence Expected

A reasoned prediction, three deliberately designed datasets, R code and output, visualisations and a defended recommendation of the most representative response-time measure.

TASK 2 – Same Average, Different Risk: Measures of Dispersion and Shape — 8 Marks Wireless Access-Point Configurations

A campus network team operates three authorized wireless access-point configurations (Config A, Config B and Config C).

All three configurations report an average client throughput close to 100 Mbps. The team manager states:

"All three configurations are performing equally well."

The security policy requires throughput to remain within ± 5% of the target for a stable, secure connection.

Task 2.1 – Predict and Diagnose | 2 Marks

Before performing any calculations, predict whether three configurations with the same average throughput are equally stable and secure.

Your response must:

- Predict whether equal means imply equal consistency and reliability.
- Explain why the mean alone fails to capture variation, distribution shape and instability.
- Identify how spread, skewness and outliers can affect connection quality and security.
- Justify why statistical analysis is necessary before accepting the manager's claim.

Task 2.2 – Design the Experimental Datasets | 2 Marks

Create or simulate three datasets in R, each containing at least 100 throughput measurements.

Your design must include:

- Config A – Low Variability: Approximately symmetric throughput with a small spread.
- Config B – High Variability: Throughput with approximately the same mean but a much larger spread.
- Config C – Distributional Irregularity: Throughput with approximately the same mean but skew or occasional dropouts (outliers).

Document the data-generation method, expected distribution characteristics and the measurements used.

Task 2.3 – Implement Statistical Analysis | 2 Marks

Write R code to calculate the following measures for all three configurations:

- Range
- Variance and standard deviation
- Interquartile range (IQR)
- Coefficient of variation (CV)
- Skewness and kurtosis using the moments or psych package
- Percentage of measurements falling outside the specified ±5% throughput limit.

Generate side-by-side boxplots and overlaid density plots to compare the three configurations.

Interpret the results and explain whether the observed differences support treating the configurations as equivalent.

Task 2.4 – Security-Stability Decision | 1 Mark

Evaluate the consistency and suitability of the three configurations.

Your analysis must:

- Identify the most informative measure of dispersion for the specified stability requirement.
- Explain when the standard deviation and the IQR could lead to different conclusions.
- Assess the significance of extreme values and distributional asymmetry.
- Recommend which configuration demonstrates the most consistent and secure output.

Task 2.5 – Changed Condition: Relative Consistency | 1 Mark

Config C now serves low-power IoT devices at a much lower target throughput, while Configs A and B continue at the original target.

Reassess the comparison using an appropriate relative measure of dispersion.

Your response must:

- Explain why the standard deviation alone is not a fair basis for comparing configurations with different target throughputs.
- Calculate and compare the coefficient of variation for all three configurations.
- Interpret the results while considering the different targets and workloads.
- Explain why relative consistency and compliance with the security policy matter more than the raw average.

Evidence Expected

A reasoned prediction, deliberately designed datasets, R code and output, comparative visualisations and a defended judgement of the most stable and secure configuration.

TASK 3 – Does One Factor Really Explain Vulnerability? Bivariate Analysis — 9 Marks

A security analytics cell has collected the following data for at least 60 authorized hosts and web applications:

Patch Age (days) | Open Ports | Configuration-Hardening Score | Failed-Login Attempts per Day | Vulnerabilities Found

The CISO has asked:

> "Which factors are related to the number of vulnerabilities found, and how strongly?"

Task 3.1 – Predict and Justify Relationships | 2 Marks

- Predict the direction (positive or negative) and approximate strength of the relationship between each factor and the number of vulnerabilities found.
- Justify each prediction using logical reasoning about security exposure and configuration.
- Record the predictions before performing any statistical analysis.
- Compare the predictions with the actual results obtained using R.

Task 3.2 – Analyse Numeric–Numeric Relationships | 3 Marks

- Construct scatter plots with fitted trend lines for each numeric factor against the vulnerability count.
- Calculate covariance and Pearson correlation coefficients to evaluate the linear relationships.
- Calculate Spearman rank correlation coefficients to examine monotonic relationships.
- Generate a correlation matrix and represent it using a heatmap or pairs plot.
- Interpret the direction, strength and nature of the relationships observed.

Task 3.3 – Compare Categorical–Numeric Relationships | 2 Marks

- Compute the group-wise mean, median and standard deviation of the vulnerability count across a categorical factor (for example the operating-system family).
- Construct grouped boxplots to compare the vulnerability distributions across the categories.
- Identify differences in central tendency, variability and potential outliers.
- Explain whether the observed group differences indicate meaningful security patterns.

Task 3.4 – Examine Categorical–Categorical Relationships | 1 Marks

- Derive a Secure/Insecure variable from the vulnerability count using a clearly stated threshold.
- Construct a two-way contingency table for firewall-present (Yes/No) against the Secure/Insecure status.
- Calculate appropriate proportions using prop.table().
- Create a stacked or grouped bar chart to compare the secure and insecure distributions.
- Interpret the association between the firewall configuration and the security outcome.

Task 3.5 – Evaluate Correlation Measures | 1 Marks

- Explain why covariance and correlation provide different types of information.
- Identify variable pairs for which the Pearson and Spearman correlation coefficients differ.
- Justify the differences using linearity, monotonicity, outliers or possible nonlinear patterns.
- Determine which correlation measure is more appropriate for each identified pair.

Evidence Expected

Reasoned predictions, R code and output for all three types of bivariate analysis, visualisations and an interpretation linking the factors to security exposure.

TASK 4 – Choosing the Right Test: One Question, Three Designs — 7 Marks

A security team is evaluating a new web-application firewall (WAF) rule set. The assessment team must compare security metrics across three study designs.

Study	Design	Question Asked
Study 1	40 web requests measured on the hardened server	Is the mean response time of the hardened server different from the 300 ms service target?

Study 2	35 requests on the new WAF rule set, 35 on the old rule set (different requests)	Does the new rule set produce a different mean blocked-attack rate from the old one?
Study 3	30 hosts measured before and after applying a hardening baseline	Does the hardening baseline reduce the vulnerability count within the same hosts?

Task 4.1 – Identify the Appropriate T-Test and Justify the Choice | 1 Marks

For each study, identify the appropriate T-test (one-sample, independent two-sample, or paired) and justify the choice based on the study design.

Task 4.2 – Formulate Hypotheses and Select the Test Direction | 2 Marks

State the null and alternative hypotheses in words and symbols for each study, and select a onesided or two-sided test with justification.

Task 4.3 – Simulate Data and Verify Assumptions | 2 Marks

Simulate realistic data for all three studies in R using set.seed() for reproducibility, and verify the normality and, where relevant, equal-variance assumptions.

Task 4.4 – Perform Statistical Tests and Interpret Results | 2 Marks

Execute the appropriate tests using t.test() in R. Report the t-statistic, degrees of freedom, pvalue and confidence interval, and interpret each result in the security context.

Evidence Expected

Correct identification of each test with justification, hypotheses, assumption checks, R code and output, and a security-focused interpretation of each result.

TASK 5 – Beyond Two Groups: One-Way ANOVA and the Multiple-Testing Problem — 7 Marks

A wireless security team is comparing the mean number of vulnerabilities found across four wireless encryption configurations (WEP, WPA, WPA2 and WPA3) on authorized lab access points.

A junior analyst proposes:

> "We will run a T-test on every pair of encryption types and pick the one with the best result."

Task 5.1 – Quantify Multiple-Testing Risk | 2 Marks

- Calculate the number of pairwise T-tests required to compare all encryption configurations.
- Calculate the probability of obtaining at least one false-positive result when many pairwise tests are performed without correction.
- Explain why conducting multiple pairwise tests without correction makes a false discovery likely.

Task 5.2 – Perform One-Way ANOVA and Identify Differences | 2 Marks

Simulate realistic vulnerability-count data for the four encryption configurations in R using set.seed() so that the results are reproducible.

Using R:

1. Present group-wise descriptive statistics and side-by-side boxplots.

2. State the null and alternative hypotheses for the One-Way ANOVA.

3. Check the normality of residuals using shapiro.test() and a Q–Q plot, and the homogeneity of variances.

4. Perform One-Way ANOVA using aov() and interpret the Sum of Squares, Mean Squares, F-statistic and p-value.

5. Perform TukeyHSD() post-hoc comparisons to identify exactly which encryption configurations differ.

6. Calculate and interpret the effect size (η^2 – eta-squared).

Task 5.3 – Investigate Assumption Violations and Make a Design Decision | 2 Marks

Modify your dataset so that one encryption group has a much larger variance, and investigate the effect on the analysis.

- Re-run the assumption checks and demonstrate that the violation is detected.
- Apply Welch's ANOVA using oneway.test() or the Kruskal–Wallis test using kruskal.test().
- Compare the conclusions with the original ANOVA and explain any differences.

The security lead must recommend one encryption configuration for the whole campus. Defend the recommendation using the statistical evidence and the security properties of each option.

Evidence Expected

A correct calculation of the multiple-testing error, assumption checks, One-Way ANOVA with post-hoc analysis and effect size, and a defended recommendation of the most secure, consistent configuration.

TASK 6– DESIGN CHALLENGE

Complete Security-Data Investigation of a Real Dataset 13 Marks

A security analyst must investigate a real-world security or network dataset to answer meaningful questions and support protection decisions, rather than simply running every function.

The dataset must contain:

- At least 300 observations
- At least three numeric variables
- At least two categorical variables, one of which must have three or more levels

Suitable sources include public network-security or traffic datasets (for example an intrusiondetection or server-log dataset) rather than trivial sample datasets.

The objective is to:

Understand and clean the data, formulate meaningful security questions, explore the data, predict outcomes, test them and defend evidence-based protection recommendations.

You are not permitted simply to run every function on every variable and report the output.

Stage 1 – Understand and Clean the Data | 2 Marks

Select one real, publicly available security dataset and use str(), summary() and appropriate R functions to understand it.

Clearly identify:

- Dataset name and source
- Meaning of the variables selected for analysis
- Variable types: nominal, ordinal, discrete and continuous
- Missing values and duplicate records
- Outliers detected using the IQR rule and/or z-scores

Justify how missing values, duplicate records and outliers are handled.

Evidence Expected

A documented data-cleaning process with appropriate R code, output and justification for each cleaning decision.

Stage 2 – Formulate Investigation Questions | 1 Mark

Develop three meaningful, domain-relevant security investigation questions using the dataset.

Your questions must correspond to:

- A T-test
- One-Way ANOVA
- A Chi-Square test

For each question, identify the response variable, explanatory variable(s) and appropriate statistical test.

Do not select a test merely because it is required. Explain why the test matches the question and variable types.

Evidence Expected

Three clearly stated investigation questions mapped to appropriate statistical tests.

Stage 3 – Perform Exploratory Data Analysis | 3 Marks

Conduct an exploratory investigation guided by your security research questions.

Your analysis must include:

- Univariate analysis: central tendency, dispersion, skewness, kurtosis, histograms and boxplots
- Bivariate analysis: correlation matrix, scatter plots, grouped boxplots and contingency tables
- At least three non-obvious observations discovered from the data

Select appropriate analyses based on the variables and questions rather than applying every technique indiscriminately.

Evidence Expected

R code, relevant statistical summaries, clearly labelled visualisations and three evidencebased observations with interpretations.

Stage 4 – Predict Before Testing | 3 Mark

Before conducting the formal statistical tests, predict whether each investigation will show a significant effect or association.

For each prediction:

- State the expected outcome.
- Identify the EDA evidence supporting the prediction.
- Explain the reasoning behind the prediction.

Do not use formal test results to justify predictions made before testing.

Evidence Expected

Three testable predictions supported by specific EDA findings.

Stage 5 – Test and Validate the Findings | 2 Marks

Conduct the three statistical investigations using R.

For each investigation:

- State the null and alternative hypotheses.
- Check the relevant assumptions.
- Select an appropriate test or alternative if assumptions are violated.
- Report the test statistic, p-value, confidence interval where applicable, and effect size.
- Perform post-hoc analysis for ANOVA where appropriate.
- Compare the actual results with the predictions made in Stage 4.

Evidence Expected

Complete R code, statistical outputs, assumption checks, effect-size calculations and interpretations for all three investigations.

Stage 6 – Defend the Findings | 2 Marks

Critically evaluate the results and prepare a final evidence-based conclusion and protection recommendation.

Your discussion must address:

- Whether each prediction was supported and why.
- The difference between statistical significance and practical significance.
- Dataset limitations, including sampling bias, missing variables and potential confounders.
- What a decision-maker can reasonably conclude from the findings.
- What conclusions cannot be justified from the analysis.

Evidence Expected

A critical evaluation of the statistical findings and a defensible security conclusion with protection recommendations.

ASSESSMENT MARK DISTRIBUTION

Question	**Assessed Area**	**Marks**

Task 1	Central Tendency – Response-Time Analysis	**6**
Task 2	Dispersion & Shape – Configuration Stability	**8**
Task 3	Bivariate Analysis – Correlation & Cross-Tabulation	**9**
Task 4	T-Tests – One-Sample, Independent & Paired	**7**
Task 5	One-Way ANOVA & Post-hoc Analysis	**7**
Task 6	Design Challenge – Complete Security-Data Investigation	**13**
	TOTAL	**50**

Submission Requirements

Submit:

- Your assignment report containing the required evidence for all tasks
- R script (.R) or R Markdown (.Rmd) files for all analyses, which must run without errors from top to bottom
- Raw and cleaned datasets used in your analysis
- All plots and tables used in your interpretation

Use set.seed() for every simulated dataset so that your results can be reproduced exactly. Assumption checks, effect sizes and written interpretation should be treated as **primary evidence**. A p-value reported without assumption checks or interpretation will not receive full credit.

Assessment Verification

Students may be asked to explain any part of their:

- Dataset design or selection
- R code
- Choice of measures and statistical tests
- Assumption checks
- Test outputs and interpretation
- Predictions
- Design Challenge

during a brief verification discussion/viva.

The student should therefore be able to explain and defend the evidence and the resulting security recommendations.

Overall Assessment Expectation

This assignment does not primarily assess your ability to run R functions.

It assesses your ability to:

Understand → Describe → Visualise → Hypothesise → Check Assumptions → Test → Interpret → Recommend Protective Measures

Your conclusions and decisions must be supported by your own analysis of authorized assessment data.

1. Expected Deliverables and Where They Appear

The assignment states what evidence is expected for every task and what must be submitted. The table below maps each expectation to the section, figure and file in this submission, so that a marker can verify completeness quickly.

Task (marks)	Expected deliverable (from the brief)	Where / file
Task 1 (6)	Reasoned prediction; three designed datasets; R code and output; histograms and boxplots; defended recommendation of the most representative response-time measure	Section 2; task1_response_times.csv; Figure 1; Screenshot 1
Task 2 (8)	Reasoned prediction; three designed datasets ($n \geq 100$); R code and output; boxplots and density plots; CV reassessment; defended choice of the most stable and secure configuration	Section 3; task2_throughput.csv, task2_c_iot.csv; Figure 2; Screenshots 2–3

Task (marks)	Expected deliverable (from the brief)	Where / file
Task 3 (9)	Predictions recorded before analysis; scatter plots, covariance, Pearson, Spearman, heatmap/pairs plot; group summaries and boxplots; contingency table, prop.table and bar chart; correlation-measure evaluation ($n \geq 60$ hosts)	Section 4; task3_hosts.csv ($n = 120$); Figures 3–7; Screenshots 4–7
Task 4 (7)	Correct t-test for each design with justification; hypotheses (words and symbols); simulated data with set.seed(); assumption checks; t.test() output with t, df, p and CI; security interpretation	Section 5; task4_study1–3.csv; Figure 8; Screenshots 8–9
Task 5 (7)	Multiple-testing calculation; descriptive statistics, boxplots, ANOVA table, residual checks, TukeyHSD, η^2; assumption violation with Welch ANOVA / Kruskal–Wallis; defended encryption recommendation	Section 6; task5_encryption*.csv; Figures 9–10; Screenshots 10–12
Task 6 (13)	Real dataset (≥ 300 rows, ≥ 3 numeric, ≥ 2 categorical incl. one with ≥ 3 levels); cleaning log; three investigation questions; EDA; predictions before testing; t-test, ANOVA and chi-square with assumptions and effect sizes; critical evaluation	Section 7; security_dataset_raw.csv, security_dataset_clean.csv; Figures 11–13; Screenshots 13–17
All	R script that runs top to bottom; raw and cleaned datasets; all plots and tables; set.seed() for each simulation; viva readiness	ITA14_Assignment4A_analysis.R; data/ folder; Appendix A; Section 8

How the work is organised. Each task section follows the same pattern required by the brief: Understand → Describe → Visualise → Hypothesise → Check assumptions → Test → Interpret → Recommend. Every section starts with the question as set, followed by the answers to each sub-task, the R code, the output screenshot, the figure and the interpretation. Predictions are always written before the results that test them.

2. Task 1 – Central Tendency Under Skewed Security Data (6 marks)

Question as set

A security team runs an authorized load and response-time assessment of a web application and records the HTTP response time (ms) of every request. Most requests complete in 200–400 ms, but a small number are much slower. The product/marketing team wants to advertise "Average response time: 320 ms." The security/operations team argues that this figure is misleading.

Tasks: 1.1 Predict and diagnose (2) · 1.2 Design the datasets (2) · 1.3 Implement and compare (1) · 1.4 Design decision and defence (1)

Task 1.1 – Predict and Diagnose (2 marks)

Prediction (recorded before any calculation). For a positively (right-) skewed response-time distribution the ordering will be **mean > median > mode**. The mode sits at the densest part of the distribution (the common 200–400 ms peak), the median is the middle request, and the mean, which weights every millisecond equally, is pulled toward the long tail of slow requests.

Distribution shape	Predicted ordering	Reason
Positively skewed (D1)	mean > median > mode	Slow tail drags the mean right; median moves less; mode ignores the tail
Approximately symmetric (D2)	mean ≈ median ≈ mode	No tail, so all three centre on the same value
Bimodal, two server nodes (D3)	mean ≈ median, both between the peaks; mode at the taller peak	The mean and median fall in the empty gap where no request actually lands

How skewness and extreme requests influence each measure

- **Mean:** it is a sum-based measure with a breakdown point of 0%. One request of 20,000 ms in 600 requests adds about 33 ms to the mean, yet it changes no typical user's experience.
- **Median:** it depends only on rank, so extreme values barely move it (breakdown point 50%). A few very slow requests change which values are ranked last, not the middle value.
- **Mode:** it ignores magnitude entirely. For continuous timings that are rounded to the millisecond it can also be unstable, because it depends on small differences in counts of single values.

Limitations of a single measure (at least two)

- **It hides the slow-tail users.** A single average says nothing about the 5% of users who wait several times longer, and these are the cases that matter for SLA breaches, time-outs, and security-relevant behaviour such as slow-loris-like degradation or overloaded authentication endpoints.
- **A skew-inflated mean overstates the typical experience** (and a mean that is advertised as "typical" misrepresents what most users see).
- **It cannot reveal multimodality.** If two servers behave differently, the single figure may describe a response time that no request ever has.

Why this is only a hypothesis. The ordering is a property of the shape of the distribution, not a law. Real traffic can be bimodal, truncated by time-outs, or contain repeated cached values, any of which can break the expected ordering. The prediction therefore has to be verified against the assessment data (Task 1.3) rather than assumed.

Task 1.2 – Design the Datasets (2 marks)

Three datasets were built in R with set.seed(14) so that the results are reproducible. All values are rounded to whole milliseconds, as a real web server log would be.

Dataset	Size	Generation method (R)	Expected statistical behaviour
D1 Positively skewed	600	round(rlnorm(600, meanlog = log(300), sdlog = 0.5))	Log-normal with median 300 ms and a realistic right tail. Theory: mean = 300·e^(0.125) ≈ 340 ms, mode = 300·e^(−0.25) ≈ 234 ms, so mean > median > mode
D2 Approximately symmetric	600	round(rnorm(600, mean = 300, sd = 40))	Normal: mean ≈ median ≈ mode ≈ 300 ms, skewness ≈ 0, no heavy tail
D3 Edge case (bimodal)	600	round(c(rnorm(300, 230, 20), rnorm(300, 520, 25)))	Two server nodes of 300 requests each, fast (≈230 ms) and slow (≈520 ms). Mean and median fall near 375 ms, an empty gap, so a single figure describes no real request

Task 1.3 – Implement and Compare (1 mark)

The mode is calculated with a self-defined function because base R's mode() returns the storage type of an object and not the statistical mode. The function counts how often each unique value occurs and returns the most frequent one.

```
if (USE_SUPPLIED_DATA) {
  t1 <- rd("task1_response_times.csv"); skewed <- t1$skewed; symm <- t1$symmetric;
bimod <- t1$bimodal
} else {
  set.seed(14)
  skewed <- round(rlnorm(600, meanlog = log(300), sdlog = 0.5))        # D1 right tail
  symm   <- round(rnorm(600, mean = 300, sd = 40))                     # D2 symmetric
  bimod  <- round(c(rnorm(300, 230, 20), rnorm(300, 520, 25)))         # D3 two nodes
}
stat <- function(x) c(mean = mean(x), median = median(x), mode = mode_fn(x),
                      trim10 = mean(x, trim = .10), trim20 = mean(x, trim = .20),
                      p95 = unname(quantile(x, .95)), sd = sd(x), skewness =
skewness(x))
print(round(sapply(list(skewed = skewed, symm = symm, bimod = bimod), stat), 2))
png("plots/t1_hist_box.png", 1200, 800, res = 150)
```

```
par(mfrow = c(2, 3))
for (nm in c("skewed", "symm", "bimod")) {
  x <- get(nm); hist(x, breaks = 30, main = nm, xlab = "Response time (ms)", col =
"steelblue")
  abline(v = c(mean(x), median(x), mode_fn(x)), lty = c(1, 2, 3), lwd = 2)
  legend("topright", c("mean", "median", "mode"), lty = 1:3, cex = .7)
}
for (nm in c("skewed", "symm", "bimod")) boxplot(get(nm), horizontal = TRUE, col =
"lightblue", xlab = "Response time (ms)")
dev.off()
```

```
> sapply(list(skewed=skewed, symm=symm, bimod=bimod), stat)   [summary table]

          skewed   symm  bimod
mean      341.51 297.93 374.43
median    296.00 297.00 367.50
mode      229.00 308.00 212.00
trim10    319.74 298.06 373.12
trim20    309.81 297.67 372.20
p95       687.15 363.05 552.00
sd        176.26  39.23 146.73
skewness    1.46  -0.08   0.02
```

Screenshot 1: Mean, median, mode, 10% and 20% trimmed mean, P95, SD and skewness for all three datasets

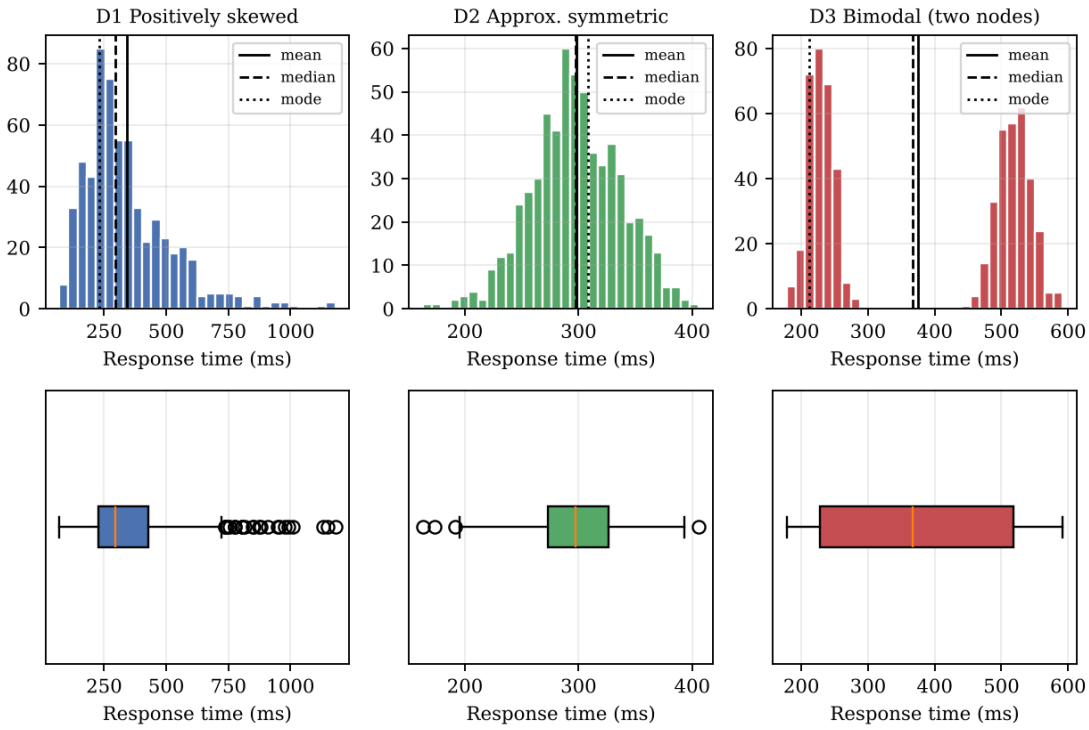


Figure 1: Histograms (with mean, median and mode marked) and boxplots of D1, D2 and D3

Measure (ms)	D1 skewed	D2 symmetric	D3 bimodal
Mean	341.5	297.9	374.4
Median	296.0	297.0	367.5
Mode	229	308	212
10% trimmed mean	319.7	298.1	373.1
20% trimmed mean	309.8	297.7	372.2
95th percentile	687.1	363.1	552.0
Skewness	1.46	-0.08	0.02

Interpretation

The prediction is confirmed for D1: mean (341.5) > median (296.0) > mode (229). The simulated values agree with the theoretical ones (mean ≈ 340, mode ≈ 234), which supports the code. The skewness of 1.46 and the boxplot, with many outliers beyond the upper whisker, show a heavy right tail. The mean is 45.5 ms (15.4%) above the median.

- **Skewness:** the 10% trimmed mean (319.7) and the 20% trimmed mean (309.8) move step by step from the mean toward the median as more of the tail is discarded. The 20% trimmed mean sits within 13.8 ms of the median, which confirms that the tail is what inflates the untrimmed mean.
- **Repeated values:** the mode of D1 (229 ms) is the single most frequent rounded value among only 600 observations; its count is small, so a different seed could move it by tens of milliseconds. In D2 the mode (308) differs from the mean (297.9) only because of the same sampling noise. The mode is therefore the least stable of the five measures for continuous timings.
- **Extreme slow requests:** they raise the mean and the SD (176.3 ms for D1 versus 39.2 for D2) but barely change the median. The 95th percentile (687 ms) is more than twice the median, which no centre measure reveals.
- **D2 (symmetric):** all measures agree within about 11 ms, so any of them is acceptable and the mean is the most efficient.
- **D3 (bimodal):** the mean (374.4) and median (367.5) land in the empty gap between the two peaks, and the trimmed means (373.1 and 372.2) do not fix it, because trimming removes tails and not a second peak. The mode (212) reports only the fast node. This is exactly the edge-case limitation that the dataset was designed to expose.

Task 1.4 – Design Decision and Defence (1 mark)

Dataset	Recommended measure	Value (ms)	Justification
D1 skewed	Median (with 20% trimmed mean as a cross-check)	296 (310)	Both resist the tail; the boxplot shows many outliers that would distort the mean
D2 symmetric	Mean	297.9	No tail, so the mean is unbiased and the most efficient; it agrees with the median
D3 bimodal	No honest single figure; report both nodes	228 and 518	The node-wise medians describe real requests; the pooled median falls in the gap

Most defensible single figure: the median (with a stated sample size and period). In the robustness comparison, the mean has a 0% breakdown point and is distorted by the slow tail; the mode is unstable and ignores magnitude; the 10%/20% trimmed means are robust to moderate tails but require an arbitrary trimming choice that a status-page reader cannot verify; the median needs no tuning and is easy to explain ("half of all requests are faster than this").

Trade-off. The median represents the usual user but hides the worst case. For D1 the median is 296 ms, while 5% of requests take longer than 687 ms. A status page that advertises only the median (or the mean) gives a false sense of reliability to the users who suffer most, and slow tails can be a symptom of resource exhaustion or an ongoing attack. The recommended publication is therefore **"median 296 ms, 95th percentile 687 ms"**, which keeps the figure honest at both ends.

On the marketing claim. The claim "average 320 ms" is a mean. On D1 the mean is 342 ms and the median 296 ms, so an average figure would overstate the typical experience by about 15% while still hiding the tail. The security/operations team's objection is correct: the mean is not a safe single number for skewed data.

Limitations of the chosen measure. The median ignores how bad the tail is (a tail of 800 ms and one of 8,000 ms give the same median); it is based on the test window only and may shift under different load; and for a multimodal system it still describes no real server. It should be published together with the percentile and the sample size.

3. Task 2 – Dispersion and Shape of Wireless Configurations (8 marks)

Question as set

A campus network team operates three authorized wireless access-point configurations (A, B and C). All three report an average client throughput close to 100 Mbps, and the manager states: "All three configurations are performing equally well." Security policy requires throughput to stay within ±5% of the target.

Tasks: 2.1 Predict and diagnose (2) · 2.2 Design the datasets (2) · 2.3 Statistical analysis (2) · 2.4 Security-stability decision (1) · 2.5 Changed condition, relative consistency (1)

Task 2.1 – Predict and Diagnose (2 marks)

Prediction (before calculation): equal means do **not** imply equal consistency, stability or security. I predict that Config A will be tight around 100 Mbps, Config B will wander widely, and Config C will have occasional dropouts, even though the three averages are almost the same.

- **Why the mean fails:** it summarises location only. Two configurations can average 100 Mbps with one at 98–102 and the other alternating between 70 and 130; the mean is identical, but the user experience is not.
- **Spread** (SD, IQR, range): high spread means clients regularly fall outside the ±5% band, producing buffering, retransmissions and time-outs.
- **Skewness and outliers:** a long lower tail (dropouts) means connections occasionally collapse. For security this matters because intermittent drops force re-authentication and reconnection, can make clients fall back to weaker or open networks, and can mask an active deauthentication or jamming attack as a "normal" quality problem.
- **Instability** also hurts monitoring: a configuration with a wide, irregular distribution makes anomaly baselines unreliable.

Why analysis is required first: the manager's claim is an empirical statement about the whole distribution of throughput, but it has been justified by one summary statistic. Spread, shape and the proportion of measurements inside the policy band must be computed before the claim can be accepted or rejected.

Task 2.2 – Design the Experimental Datasets (2 marks)

Config	n	Generation (R, set.seed(24))	Expected characteristics
A – low variability	120	rnorm(120, mean = 100, sd = 2)	Symmetric, small spread, CV ≈ 2%, nearly all values within 95–105 Mbps
B – high variability	120	rnorm(120, mean = 100, sd = 10)	Symmetric but wide, CV ≈ 10%, a large share of values outside the ±5% band

Config	n	Generation (R, set.seed(24))	Expected characteristics
C – irregular	120	c(rnorm(110, 100, 3), rnorm(10, 60, 5))	Tight core with 10 dropouts near 60 Mbps: strong negative skew, high kurtosis, mean pulled slightly below 100

Measurements: throughput in Mbps per client sample. The policy band is 95–105 Mbps (±5% of the 100 Mbps target). The mean of the simulated data is close to, but not exactly, 100, because sampling noise (and, for C, the dropouts) shifts it; the requirement is “close to 100”, which is met by all three.

Task 2.3 – Implement Statistical Analysis (2 marks)

```
if (USE_SUPPLIED_DATA) {
  t2 <- rd("task2_throughput.csv"); A <- t2$A; B <- t2$B; C <- t2$C; Ciot <-
rd("task2_c_iot.csv")$C_iot
} else {
  set.seed(24)
  A <- rnorm(120, 100, 2); B <- rnorm(120, 100, 10); C <- c(rnorm(110, 100, 3),
rnorm(10, 60, 5))
  set.seed(25); Ciot <- c(rnorm(110, 10, 0.3), rnorm(10, 6, 0.5))
}
desc <- function(x, target = 100) c(mean = mean(x), range = diff(range(x)), var =
var(x), sd = sd(x),
  IQR = IQR(x), CV_pct = sd(x) / mean(x) * 100, skewness = skewness(x), kurtosis =
kurtosis(x),
  pct_outside_5 = mean(abs(x - target) / target > .05) * 100)
print(round(sapply(list(A = A, B = B, C = C), desc), 2))
png("plots/t2_box_density.png", 1200, 560, res = 150); par(mfrow = c(1, 2))
boxplot(A, B, C, names = c("A", "B", "C"), ylab = "Throughput (Mbps)", col =
"lightblue"); abline(h = c(95, 105), col = "red", lty = 2)
plot(density(A), xlim = c(40, 140), main = "Overlaid densities", xlab = "Throughput
(Mbps)", col = "blue", lwd = 2)
lines(density(B), col = "orange", lwd = 2); lines(density(C), col = "darkgreen", lwd =
2); abline(v = c(95, 105), col = "red", lty = 2)
legend("topleft", c("A", "B", "C"), col = c("blue", "orange", "darkgreen"), lwd = 2);
dev.off()
# 2.5 relative consistency (C now an IoT config with target 10 Mbps)
t25 <- rbind(A = desc(A, 100)[c("mean", "sd", "CV_pct", "pct_outside_5")],
             B = desc(B, 100)[c("mean", "sd", "CV_pct", "pct_outside_5")],
             C_IoT = desc(Ciot, 10)[c("mean", "sd", "CV_pct", "pct_outside_5")])
print(round(t25, 2))
```

```
> sapply(list(A=A,B=B,C=C), desc)

                A      B      C
mean        99.68  97.74  96.75
range       11.10  52.68  51.24
var          4.46 100.09 126.30
sd           2.11  10.00  11.24
IQR          2.81  12.67   4.43
CV_pct       2.12  10.24  11.62
skewness     0.04  -0.34  -2.77
kurtosis     2.86   3.37   9.62
pct_outside_5 3.33  55.83  17.50
```

Screenshot 2: Dispersion and shape measures for Configs A, B and C (CV in %, kurtosis non-excess, as in the moments package)

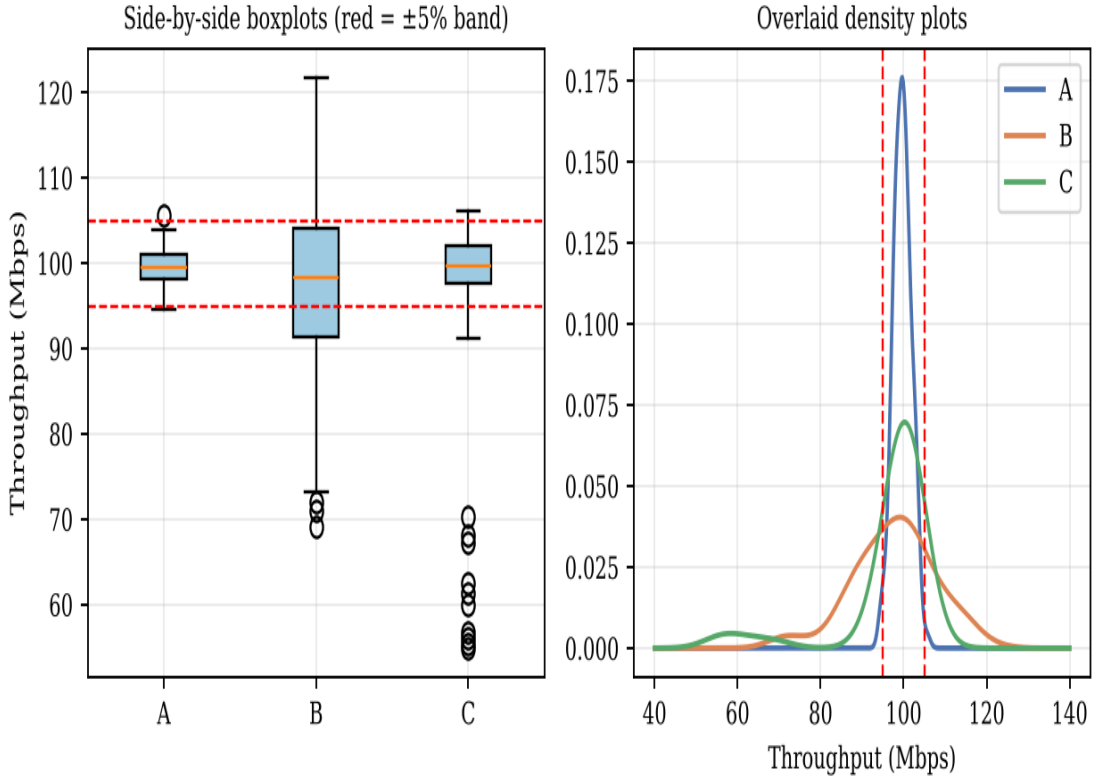


Figure 2: Side-by-side boxplots (left) and overlaid density plots (right); red dashed lines mark the ±5% band

Measure	Config A	Config B	Config C
Mean (Mbps)	99.68	97.74	96.75
Range	11.10	52.68	51.24
Variance	4.46	100.09	126.30
Standard deviation	2.11	10.00	11.24
IQR	2.81	12.67	4.43
CV (%)	2.12	10.24	11.62
Skewness	0.04	-0.34	-2.77
Kurtosis	2.86	3.37	9.62
% outside ±5%	3.3	55.8	17.5

Interpretation

- **Config A:** SD 2.11, CV 2.12%, skewness 0.04 and kurtosis 2.86 (normal-like, the normal value is 3). 3.3% of measurements fall outside the band.
- **Config B:** symmetric but with SD 10.00 and CV 10.24%, about five times the spread of A. **55.8%** of its measurements violate the ±5% policy, so more than half of the time the connection is outside the stable range.
- **Config C:** its IQR (4.43) is small, so the typical measurement is stable, but the dropouts create skewness -2.77, kurtosis 9.62 (far above 3, meaning heavy tails), a range of 51.2 Mbps and an SD of 11.24, larger than B's. 17.5% of measurements are outside the band, and the mean (96.75) is pulled below the target by the dropouts.

Do the differences support treating the configurations as equivalent? No. Equal averages hide a 5-fold difference in spread (A versus B), a heavy lower tail (C), and compliance rates that differ from about 3% to about 56% outside the band. The manager's statement is rejected.

Task 2.4 – Security-Stability Decision (1 mark)

Most informative dispersion measure for the stated requirement. The policy is expressed as a percentage band, so the most informative quantity is the **percentage of measurements outside ±5%** (a direct compliance rate), supported by the **coefficient of variation**, which expresses the spread relative to the target. Range is too sensitive to a single value, and variance is in squared units.

When SD and IQR lead to different conclusions. SD squares deviations, so it reacts strongly to outliers; the IQR ignores the outer 50% of the data. Config C shows this: SD = 11.24 (larger than B's 10.00) suggests C is as unstable as B, whereas IQR = 4.43 (less than half of B's 12.67) says that its core is tight. Neither is wrong; they answer different questions. SD answers "how much does it vary, dropouts included?"; the IQR answers "how stable is the typical sample?". Because security cares about dropouts, the IQR alone would understate the risk in C, and SD alone would overstate its day-to-day variability. Both should be reported.

Extreme values and asymmetry. In C the dropouts are rare (about 8% of measurements) but each one is a service interruption; skewness of -2.77 and kurtosis of 9.62 flag a risk that the mean and IQR cannot see.

Config	% outside ±5%	CV (%)	Verdict (assuming ≥ 95% of samples must be in band)
A	3.3	2.12	Compliant (about 97% in band); most consistent
B	55.8	10.24	Not compliant: wide, unstable
C	17.5	11.62	Not compliant: dropouts, heavy lower tail

Recommendation: Config A is the most consistent and policy-compliant. (The 95% compliance threshold is my working assumption, because the brief gives only the ±5% band; the ranking A > C > B holds for any reasonable threshold.) Config C should not be rolled out before the cause of the dropouts (interference, roaming failure, or an attack such as deauthentication) is investigated.

Task 2.5 – Changed Condition: Relative Consistency (1 mark)

Config C now serves low-power IoT devices with a target of 10 Mbps (simulated with set.seed(25) as 110 values from N(10, 0.3) plus 10 dropouts from N(6, 0.5)); A and B remain at 100 Mbps.

Why SD alone is not a fair basis. The SD is in the units of the data, so it scales with the target: an SD of 1.2 Mbps is huge on a 10 Mbps link (12%) and trivial on a 100 Mbps link (1.2%). Comparing raw SDs across different targets rewards the configuration that simply runs at a lower speed.

```
> data.frame(mean, sd, CV%, %outside +/-5%) per configuration

             A (target 100)  B (target 100)  C-IoT (target 10)
mean                  99.68           97.74               9.68
sd                     2.11           10.00               1.23
CV_pct                 2.12           10.24              12.74
pct_outside_5          3.33           55.83              15.83
```

Screenshot 3: Mean, SD, CV and % outside ±5% of the respective target (C is now the IoT configuration with a 10 Mbps target)

Measure	A (100 Mbps)	B (100 Mbps)	C-IoT (10 Mbps)
Mean	99.68	97.74	9.68
Standard deviation	2.11	10.00	1.23

Measure	A (100 Mbps)	B (100 Mbps)	C-IoT (10 Mbps)
CV (%)	2.12	10.24	12.74
% outside ±5% of own target	3.3	55.8	15.8

Interpretation. By raw SD, C-IoT (1.23) looks even *more* stable than A (2.11), which would be a wrong conclusion. By CV, C-IoT is the *least* consistent (12.74%), above B (10.24%) and far above A (2.12%), and 15.8% of its measurements are outside its own ±5% band. The order of consistency is A > B > C-IoT.

Why relative consistency and compliance matter more than the raw average. The raw average says nothing about whether the link behaves predictably, and IoT devices are often the least able to recover from a dropout (limited buffering, re-association costs battery, and failed re-authentication can leave devices offline or on insecure fallbacks). The security policy is written as a relative tolerance (±5% of target), so CV and band compliance are the measures that match it. **Caveat:** CV is only meaningful for ratio-scale data with a mean well above zero, and the dropouts inflate it, so CV should always be reported with the percentage outside the band.

4. Task 3 – Bivariate Analysis of Vulnerability Factors (9 marks)

Question as set

A security analytics cell has collected data for at least 60 authorized hosts and web applications: patch age (days), open ports, configuration-hardening score, failed-login attempts per day and vulnerabilities found. The CISO asks: “Which factors are related to the number of vulnerabilities found, and how strongly?”

Tasks: 3.1 Predict and justify (2) · 3.2 Numeric–numeric (3) · 3.3 Categorical–numeric (2) · 3.4 Categorical–categorical (1) · 3.5 Evaluate correlation measures (1)

Data. 120 authorized hosts were simulated with set.seed(3) (file task3_hosts.csv). The variables are patch_age (days), open_ports (count), hardening (score 0–100), failed_logins (per day), vulns (count of vulnerabilities found), os_family (Windows, Linux, macOS) and firewall (Yes/No). The vulnerability count was generated from a Poisson model in which exposure raises the count and hardening and a firewall lower it. The failed-login variable was deliberately

generated without a causal link to vulnerabilities, and a few extreme values were added, as in real logs.

Task 3.1 – Predict and Justify Relationships (2 marks)

The predictions below were written before any statistic was computed.

Factor	Predicted direction and strength	Justification
Patch age	Positive, strong (r ≈ +0.6 to +0.8)	Each missed patch cycle leaves publicly known vulnerabilities open, so older systems accumulate findings.
Open ports	Positive, moderate (r ≈ +0.4 to +0.6)	Each listening service is an additional attack surface and a possible vulnerable service version.
Hardening score	Negative, strong (r ≈ −0.6 to −0.8)	Hardening disables unneeded services and enforces secure settings, which removes findings.
Failed logins per day	Weak or none (|r| < 0.2)	Failed logins reflect attacker activity or user error, which is a symptom of being targeted and not a cause of vulnerabilities.

Task 3.2 – Numeric–Numeric Relationships (3 marks)

```
if (USE_SUPPLIED_DATA) {
  df <- rd("task3_hosts.csv")
} else {
  set.seed(3); n <- 120
  patch <- round(rgamma(n, 2, scale = 45)); ports <- rpois(n, 5 + patch / 45)
  hard <- round(pmin(pmax(rnorm(n, 72 - patch * .07, 11), 15), 100), 1)
  fl <- rpois(n, 9); fl[sample(n, 6)] <- sample(90:159, 6, TRUE)
  os <- sample(c("Windows", "Linux", "macOS"), n, TRUE, c(.45, .40, .15)); fw <-
sample(c("Yes", "No"), n, TRUE, c(.6, .4))
  lam <- exp(.55 + .0075 * patch + .085 * ports - .015 * hard + .40 * (os ==
"Windows") - .25 * (os == "macOS") - .55 * (fw == "Yes"))
  df <- data.frame(host = sprintf("H%03d", 1:n), patch_age = patch, open_ports =
ports, hardening = hard,
```

```
                   failed_logins = fl, os_family = os, firewall = fw, vulns = rpois(n,
lam))
}
num <- c("patch_age", "open_ports", "hardening", "failed_logins")
# 3.1 predictions (recorded in the report BEFORE this block): patch_age +strong,
open_ports +moderate,
#      hardening -strong, failed_logins ~0
res3 <- data.frame(
  covariance = sapply(num, function(v) cov(df[[v]], df$vulns)),
  pearson_r  = sapply(num, function(v) cor(df[[v]], df$vulns)),
  pearson_p  = sapply(num, function(v) cor.test(df[[v]], df$vulns)$p.value),
  spearman_rho = sapply(num, function(v) cor(df[[v]], df$vulns, method = "spearman")),
  spearman_p = sapply(num, function(v) suppressWarnings(cor.test(df[[v]], df$vulns,
method = "spearman"))$p.value))
print(round(res3, 4))
print(round(cor(df[c(num, "vulns")]), 3))
png("plots/t3_scatter.png", 1200, 1000, res = 150); par(mfrow = c(2, 2))
for (v in num) { plot(df[[v]], df$vulns, xlab = v, ylab = "vulns", pch = 19, col =
rgb(0, 0, 1, .5)); abline(lm(df$vulns ~ df[[v]]), col = "red", lwd = 2) }
dev.off()
png("plots/t3_pairs.png", 1000, 1000, res = 150); pairs(df[c(num, "vulns")], pch = 19,
col = rgb(0, 0, 1, .4)); dev.off()
cm <- cor(df[c(num, "vulns")])
png("plots/t3_heatmap.png", 800, 700, res = 150); heatmap(cm, Rowv = NA, Colv = NA,
symm = TRUE, scale = "none", col = hcl.colors(20, "RdBu", rev = TRUE), margins = c(9,
9)); dev.off()
print(aggregate(vulns ~ os_family, df, function(x) c(n = length(x), mean = mean(x),
median = median(x), sd = sd(x), max = max(x))))
png("plots/t3_osbox.png", 800, 600, res = 150); boxplot(vulns ~ os_family, df, col =
"lightblue", ylab = "Vulnerabilities"); dev.off()
print(kruskal.test(vulns ~ os_family, df))
df$status <- ifelse(df$vulns >= 5, "Insecure", "Secure")     # stated threshold: >= 5
vulnerabilities
tab <- table(Firewall = df$firewall, Status = df$status); print(tab);
print(round(prop.table(tab, 1), 3))
print(chisq.test(tab, correct = FALSE))
png("plots/t3_bars.png", 1200, 560, res = 150); par(mfrow = c(1, 2))
barplot(tab, beside = TRUE, legend = TRUE, main = "Counts", col = c("tomato",
"seagreen"))
barplot(t(prop.table(tab, 1)) * 100, legend = TRUE, main = "% within firewall group",
col = c("tomato", "seagreen")); dev.off()
ok <- df$failed_logins < 60   # sensitivity: remove the extreme failed-login hosts
print(c(pearson = cor(df$failed_logins[ok], df$vulns[ok]), spearman =
cor(df$failed_logins[ok], df$vulns[ok], method = "spearman")))
```

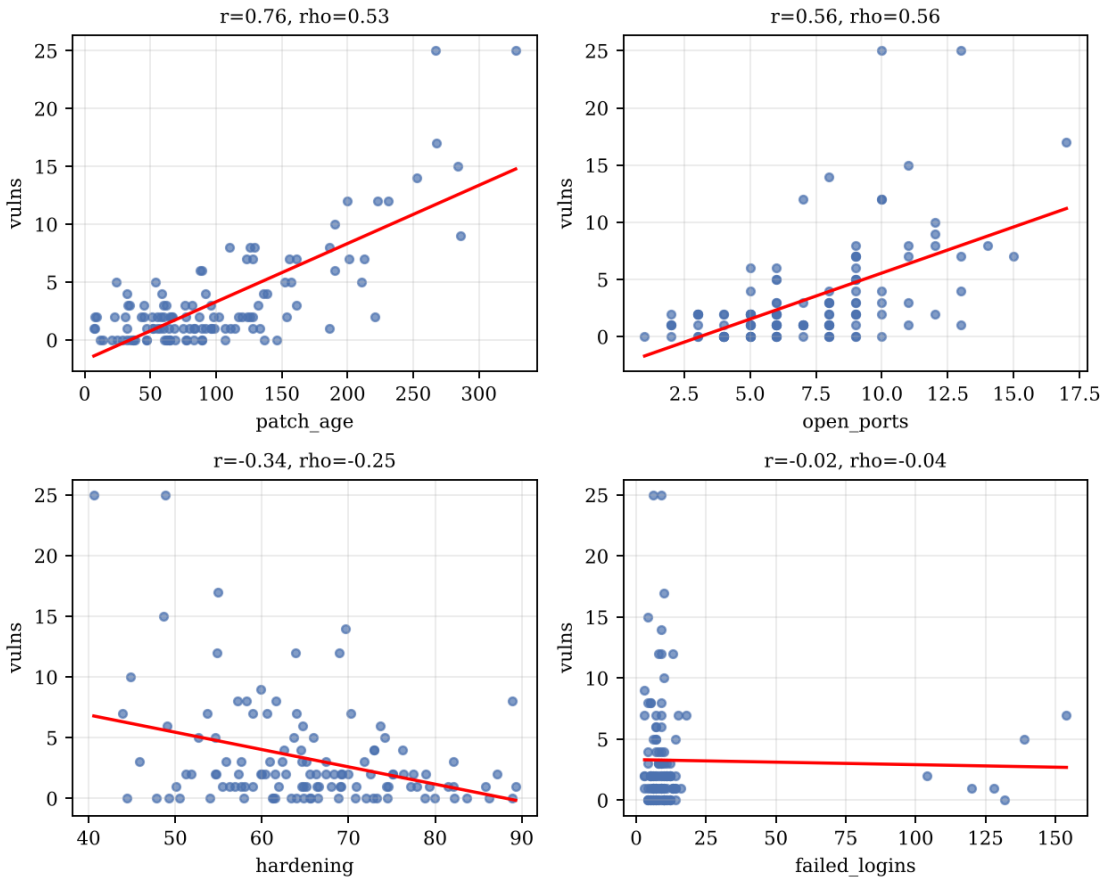


Figure 3: Scatter plots of vulnerabilities against each numeric factor with least-squares trend lines (r and ρ in each title)

```
> cor / cov / cor(method='spearman') with vulns

               covariance  pearson_r  pearson_p  spearman_rho  spearman_p
patch_age         229.0235     0.7612     0.0000        0.5340      0.0000
open_ports          7.6426     0.5564     0.0000        0.5603      0.0000
hardening         -16.4682    -0.3436     0.0001       -0.2543      0.0051
failed_logins      -2.9765    -0.0248     0.7880       -0.0408      0.6583
```

Screenshot 4: Covariance, Pearson r and Spearman ρ (with p-values) between each factor and vulns

```
> round(cor(df[num_vars]), 3)

              patch_age  open_ports  hardening  failed_logins  vulns
patch_age         1.000       0.545     -0.328          0.022  0.761
open_ports        0.545       1.000     -0.117          0.070  0.556
hardening        -0.328      -0.117      1.000         -0.088 -0.344
failed_logins     0.022       0.070     -0.088          1.000 -0.025
vulns             0.761       0.556     -0.344         -0.025  1.000
```

Screenshot 5: Pearson correlation matrix of the numeric variables

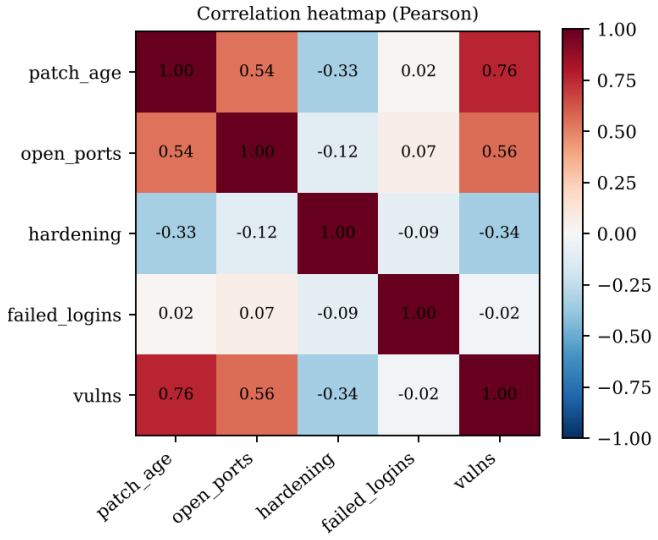


Figure 4: Correlation heatmap

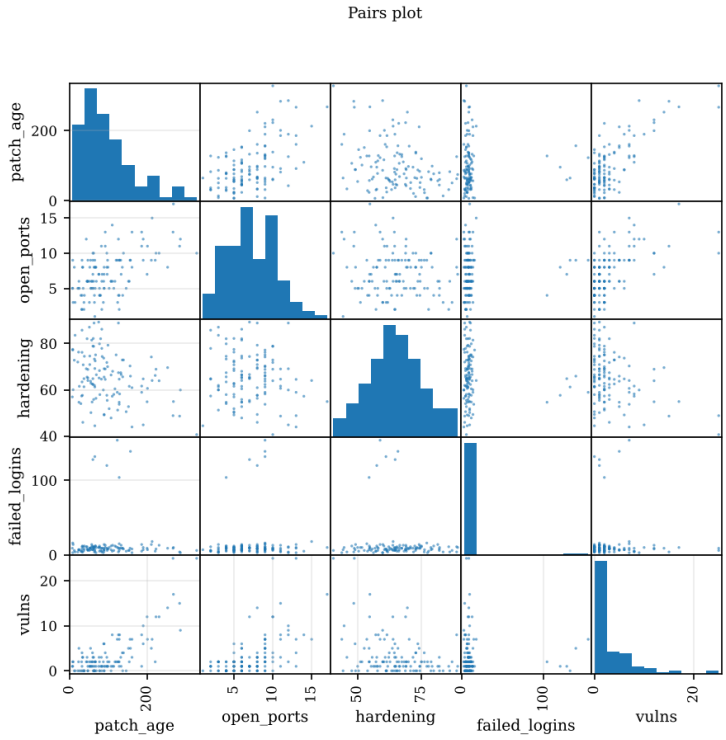


Figure 5: Pairs plot of the four factors and vulnerabilities

Factor	Covariance	Pearson r (p)	Spearman ρ (p)	Predicted	Result
patch_age	229.02	0.761 (< 2.2e-16)	0.534 (< 2.2e-16)	+ strong	Confirmed

Factor	Covariance	Pearson r (p)	Spearman ρ (p)	Predicted	Result
open_ports	7.64	0.556 (< 2.2e-16)	0.560 (< 2.2e-16)	+ moderate	Confirmed
hardening	-16.47	-0.344 (1.00e-4)	-0.254 (0.0051)	− strong	Direction right, weaker
failed_logins	-2.98	-0.025 (0.7880)	-0.041 (0.6583)	≈ 0	Confirmed

Comparison of predictions with the actual results

- **Patch age (r = 0.76, ρ = 0.53):** strong positive, as predicted. Older patch levels go with more vulnerabilities.
- **Open ports (r = 0.56, ρ = 0.56):** moderate positive, as predicted. The two coefficients agree, which indicates a consistent monotonic pattern.
- **Hardening score (r = -0.34, ρ = -0.25):** the direction is right (negative) and significant, but the strength is lower than predicted. A likely reason is that hardening and patch age are themselves related (r = -0.33), and patch age carries most of the explanatory weight. The prediction of "strong" was too optimistic, which is a useful lesson: a plausible mechanism does not guarantee a strong correlation.
- **Failed logins (r = -0.02, p = 0.79):** no relationship, as predicted. This is itself a finding: monitoring failed logins is useful for detecting attacks, but it is not an indicator of how vulnerable a host is.

Correlation is not causation. These coefficients measure association between observed variables; patch age, open ports and hardening are intercorrelated, so a multivariable model (for example Poisson regression) would be needed to separate their individual effects.

Task 3.3 – Categorical–Numeric Relationships (2 marks)

```
> aggregate(vulns ~ os_family, df, ...)

           count  mean  median  std  max
os_family
Linux         50  2.34    1.00 3.43   17
Windows       52  4.60    2.00 5.45   25
macOS         18  2.00    1.00 2.61    9
```

Screenshot 6: Group-wise count, mean, median, SD and maximum of vulnerabilities by operating-system family

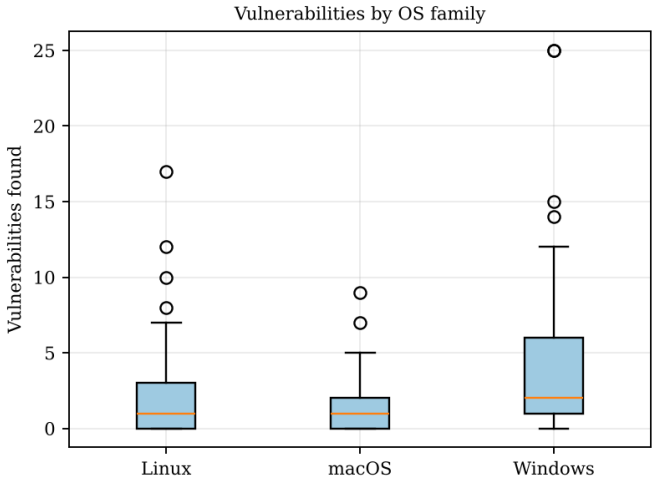


Figure 6: Grouped boxplots of vulnerabilities by OS family

OS family	n	Mean	Median	SD	Max
Linux	50	2.34	1.0	3.43	17
macOS	18	2.00	1.0	2.61	9
Windows	52	4.60	2.0	5.45	25

Central tendency: Windows hosts have the highest mean (4.60) and median (2.0); Linux (2.34) and macOS (2.00) are about half as high. **Variability:** Windows also has the largest SD (5.45), meaning that its hosts differ most from each other. **Outliers:** each group has a right-skewed distribution with a few hosts with many findings (Windows up to 25). Means are larger than medians in every group because of this skew, so the medians are the safer summary.

Are the differences meaningful? A Kruskal–Wallis test (suitable for skewed counts) gives $\chi^2 = 12.80$, $p = 0.0017$, so the difference between OS families is unlikely to be chance. The pattern

is a useful hint for prioritising hardening and patch effort on Windows hosts, but it should not be read as "Windows is less secure by nature": the macOS group is small (n = 18), and OS family is confounded with patching practice, role and exposure. It is a pattern to investigate, not proof of cause.

Task 3.4 – Categorical–Categorical Relationships (1 mark)

Derived variable. status = "Insecure" if vulns ≥ 5, otherwise "Secure". The threshold of 5 findings is stated as a working policy value (a host with five or more findings is treated as needing remediation). Different thresholds will change the proportions, so the conclusion should be checked for sensitivity.

```
> tab <- table(dffirewall,dfstatus); prop.table(tab,1)

Counts:
status    Insecure  Secure
firewall
No              19      32
Yes              9      60

prop.table(tab, 1):
status    Insecure  Secure
firewall
No           0.373   0.627
Yes          0.130   0.870

chisq.test(tab, correct=FALSE): X-squared = 9.609, df = 1, p-value = 0.0019
```

Screenshot 7: Contingency table, row proportions from prop.table() and chi-square test for firewall × status

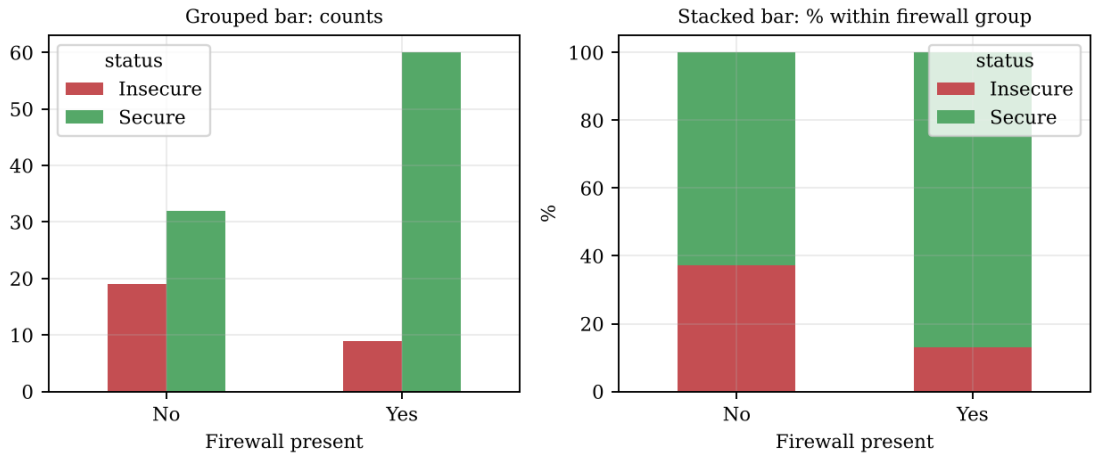


Figure 7: Grouped (counts) and stacked (percentage) bar charts of secure and insecure hosts by firewall presence

Firewall	Secure	Insecure	% Insecure
No	32	19	37.3
Yes	60	9	13.0

Interpretation. 37.3% of hosts without a firewall are insecure, against 13.0% with a firewall. A chi-square test of independence gives $\chi^2 = 9.61$, df = 1, p = 0.0019, so the association is statistically significant. The direction is consistent with a firewall reducing exposure. It is an **association, not proof of causation**: hosts that have firewalls may also be better managed in other respects.

Task 3.5 – Evaluate Correlation Measures (1 mark)

Covariance versus correlation. Covariance shows the **direction** of the joint variation but its magnitude depends on the units and variances of the variables; for example, the covariance of patch age and vulnerabilities is 229.0 (days × findings), while open ports gives 7.6, and these two numbers cannot be compared with each other. Correlation divides by both standard deviations, yielding a unit-free value in [−1, 1] that expresses **strength** and allows comparison across variable pairs.

Pair	Pearson r	Spearman ρ	Difference	More appropriate
patch_age – vulns	0.761	0.534	0.227	Spearman
hardening – vulns	-0.344	-0.254	-0.089	Spearman
open_ports – vulns	0.556	0.560	-0.004	Either (they agree)
failed_logins – vulns	-0.025	-0.041	0.016	Spearman (outliers)

- **patch_age – vulns:** Pearson (0.76) is clearly larger than Spearman (0.53). Vulnerability counts rise faster than linearly for the oldest, most exposed hosts, so a few high-leverage points inflate Pearson, while the rank-based Spearman describes the monotonic trend. The relationship is monotonic but not linear, so **Spearman** is the more appropriate summary (a Poisson or log-linear model would be better still).
- **hardening – vulns:** Pearson (-0.34) is a little stronger than Spearman (-0.25). The score is clipped at its limits and the counts are skewed, so **Spearman** is safer.

- **open_ports – vulns:** the coefficients are close (0.56 and 0.56), so the pattern is both monotonic and approximately linear.
- **failed_logins – vulns:** both are about zero. The variable has extreme values (up to 154 failed logins per day against a typical 9); removing hosts with 60 or more changes Pearson from -0.025 to -0.046 and Spearman from -0.041 to -0.044, so the conclusion "no relationship" does not depend on the outliers. Spearman is preferred because of them.

5. Task 4 – Choosing the Right T-Test: One Question, Three Designs (7 marks)

Question as set

A security team evaluates a new web-application firewall (WAF) rule set across three study designs: **Study 1** – 40 web requests on the hardened server; is the mean response time different from the 300 ms service target? **Study 2** – 35 requests on the new rule set and 35 different requests on the old rule set; does the new rule set produce a different mean blocked-attack rate? **Study 3** – 30 hosts measured before and after a hardening baseline; does the baseline reduce the vulnerability count within the same hosts?

Tasks: 4.1 Identify the t-test (1) · 4.2 Hypotheses and direction (2) · 4.3 Simulate and verify assumptions (2) · 4.4 Perform the tests and interpret (2)

Task 4.1 – Identify the Appropriate T-Test (1 mark)

Study	Design	Appropriate test	Justification
1	One sample of 40 requests compared with a fixed benchmark (300 ms)	**One-sample t-test**	There is one group and a known constant; the question is whether the population mean equals that constant
2	Two groups of different requests (35 versus 35)	**Independent two-sample t-test** (Welch version unless variances are shown equal)	The observations in the two groups are unrelated; each request belongs to only one rule set

Study	Design	Appropriate test	Justification
3	The same 30 hosts measured twice	**Paired t-test**	Measurements are dependent (same host before and after). Pairing removes between-host variation and tests the mean of the differences

Task 4.2 – Hypotheses and Test Direction (2 marks)

Study	H0 (words and symbols)	H1 (words and symbols)	Direction and justification
1	The mean response time of the hardened server equals the service target. H0: $\mu = 300$	The mean differs from the target. H1: $\mu \neq 300$	**Two-sided.** The question asks whether it is "different"; a mean above target (service degradation) and below target (possibly over-provisioned or cached) both matter
2	The new and old rule sets have the same mean blocked-attack rate. H0: $\mu_new = \mu_old$	The means differ. H1: $\mu_new \neq \mu_old$	**Two-sided.** The question is "different", and a new rule set could also block less (a security regression), which a one-sided test would not detect
3	The baseline does not change the mean vulnerability count. H0: $\mu_d = 0$, with d = before − after	The baseline reduces the count. H1: $\mu_d > 0$	**One-sided (greater).** The question is directional ("reduce") and the hardening is expected to help; a result in the opposite direction would be treated as "no reduction"

The significance level is $\alpha = 0.05$ for all studies.

Task 4.3 – Simulate Data and Verify Assumptions (2 marks)

All data were simulated with set.seed(4). Study 1: N(312, 35), n = 40 (a hardened server slightly above target). Study 2: new rule set N(86, 6) and old rule set N(82, 7) percent blocked, n = 35 each. Study 3: before ~ N(14, 3.2) vulnerabilities per host, after = max(before − N(2.5, 2.2), 0), n = 30.

```
if (USE_SUPPLIED_DATA) {
  s1 <- rd("task4_study1.csv")$study1_resp_ms; s2 <- rd("task4_study2.csv"); w_new <-
s2$new_rate; w_old <- s2$old_rate
  s3 <- rd("task4_study3.csv"); bef <- s3$before; aft <- s3$after
} else {
  set.seed(4)
  s1 <- rnorm(40, 312, 35); w_new <- rnorm(35, 86, 6); w_old <- rnorm(35, 82, 7)
  bef <- rnorm(30, 14, 3.2); aft <- pmax(bef - rnorm(30, 2.5, 2.2), 0)
}
dd <- bef - aft
print(shapiro.test(s1)); print(shapiro.test(w_new)); print(shapiro.test(w_old));
print(var.test(w_new, w_old)); print(shapiro.test(dd))
print(t.test(s1, mu = 300, alternative = "two.sided"))                        # Study 1
print(t.test(w_new, w_old, var.equal = FALSE, alternative = "two.sided"))     # Study 2
(Welch)
print(t.test(bef, aft, paired = TRUE, alternative = "greater"))               # Study 3
(d = before - after)
print(wilcox.test(bef, aft, paired = TRUE, alternative = "greater"))          #
sensitivity check (differences borderline-normal)
png("plots/t4_qq.png", 1400, 450, res = 150); par(mfrow = c(1, 4))
for (x in list(s1, w_new, w_old, dd)) { qqnorm(x, main = ""); qqline(x, col = "red")
}; dev.off()
```

```
> Assumption checks

Study 1  shapiro.test(resp_ms):   p = 0.830
Study 2  shapiro.test(new_rate):  p = 0.144
Study 2  shapiro.test(old_rate):  p = 0.704
Study 2  var.test(new, old):      F = 0.612, p = 0.158
Study 3  shapiro.test(before-after): p = 0.058
```

Screenshot 8: Assumption checks: Shapiro–Wilk for each sample, F-test for equal variances (Study 2), Shapiro–Wilk of the paired differences (Study 3)

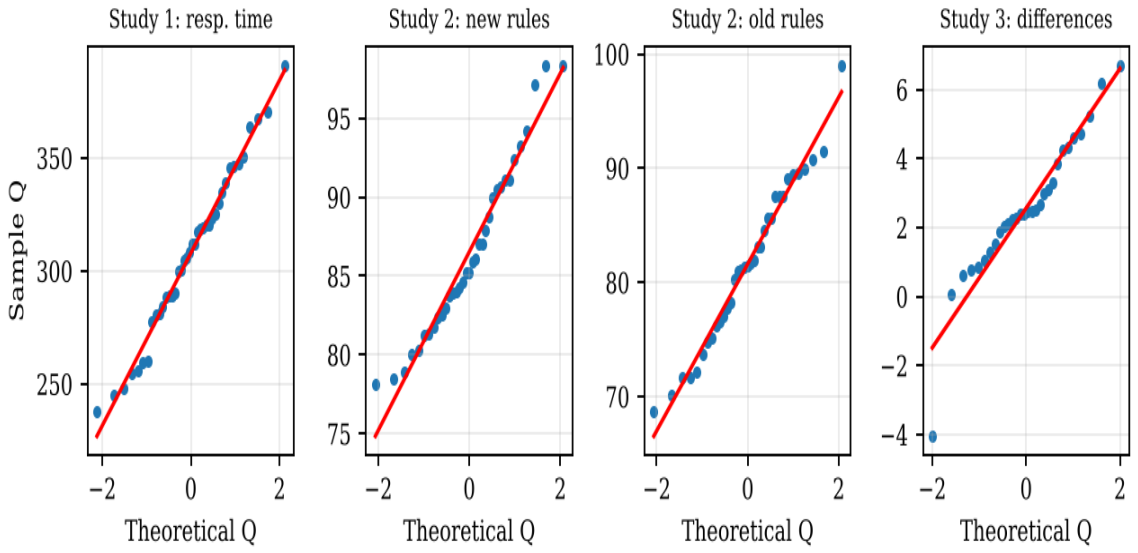


Figure 8: Normal Q–Q plots for Study 1, the two Study 2 groups and the Study 3 differences

Check	Result	Conclusion
Study 1 normality (Shapiro–Wilk)	p = 0.830	No evidence against normality
Study 2 normality, new / old	p = 0.144 / 0.704	No evidence against normality in either group
Study 2 equal variances (F-test)	F = 0.612, p = 0.158	No evidence of unequal variances; Welch is still used as a safe default
Study 3 normality of differences	p = 0.058	Borderline; with n = 30 the t-test is robust, and a Wilcoxon signed-rank test is run as a sensitivity check

Independence of observations is assumed by design (separate requests; different hosts). The Q–Q plots lie close to the reference line apart from slight tail deviations.

Task 4.4 – Perform the Tests and Interpret Results (2 marks)

```
> t.test(...) for Studies 1-3

One Sample t-test (Study 1)   data: resp_ms
t = 1.334, df = 39, p-value = 0.1899
alternative hypothesis: true mean is not equal to 300
95 percent confidence interval: 295.94  319.78
sample estimates: mean of x = 307.86

Welch Two Sample t-test (Study 2)   data: new_rate and old_rate
t = 3.216, df = 64.29, p-value = 0.002036
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval: 1.86  7.97
sample estimates: mean of x = 86.53, mean of y = 81.61

Paired t-test (Study 3)   data: before and after
t = 6.883, df = 29, p-value = 7.292e-08
alternative hypothesis: true mean difference is greater than 0
95 percent confidence interval: 1.928  Inf
sample estimates: mean difference (before-after) = 2.559
```

Screenshot 9: t.test() output for Study 1 (one-sample), Study 2 (Welch two-sample) and Study 3 (paired, one-sided)

Study	Test	t	df	p-value	95% CI	Effect size
1	One-sample, two-sided	1.334	39	0.1899	[295.94, 319.78]	d = 0.21
2	Welch, two-sided	3.216	64.29	0.0020	[1.86, 7.97]	d = 0.77
3	Paired, one-sided (greater)	6.883	29	7.29e-8	[1.93, ∞)	dz = 1.26

- **Study 1 – fail to reject H0.** The sample mean is 307.9 ms, which is 7.9 ms above target, but the 95% CI [295.9, 319.8] contains 300 and p = 0.19 is above 0.05. There is no statistical evidence that the hardened server's mean differs from the 300 ms target. This is not proof that it equals 300: with n = 40, differences of about ±12 ms cannot be excluded.
- **Study 2 – reject H0.** The new rule set blocks 86.5% of attacks on average against 81.6% for the old set, a difference of 4.9 percentage points (Welch t = 3.22, df = 64.3, p = 0.0020). The 95% CI [1.9, 8.0] excludes 0 and the effect is medium to large (d = 0.77). The new rule set blocks more attacks, so replacing the old one is justified,

although a test on production traffic and a check of false-positive rates (legitimate requests blocked) are needed before deployment.

- **Study 3 – reject H0.** The baseline reduced the mean vulnerability count by 2.56 per host (from 14.91 to 12.35); paired t(29) = 6.88, p = 7.29e-8, and the one-sided 95% CI lower bound is 1.93, so the reduction is at least about 1.9 vulnerabilities per host. The effect is large (dz = 1.26). The Wilcoxon signed-rank sensitivity test agrees (p = 5.96e-7). Because the design is before/after without a control group, the result shows that the counts fell, but cannot exclude other contributing changes during the period.

6. Task 5 – One-Way ANOVA and the Multiple-Testing Problem (7 marks)

Question as set

A wireless security team compares the mean number of vulnerabilities across four encryption configurations (WEP, WPA, WPA2, WPA3) on authorized lab access points. A junior analyst proposes: "We will run a t-test on every pair of encryption types and pick the one with the best result."

Tasks: 5.1 Quantify multiple-testing risk (2) · 5.2 One-way ANOVA and post-hoc (2) · 5.3 Assumption violation and design decision (2, plus the recommendation)

Task 5.1 – Quantify Multiple-Testing Risk (2 marks)

Number of pairwise tests. For k = 4 groups the number of pairs is C(4, 2) = 4 × 3 / 2 = **6** (WEP–WPA, WEP–WPA2, WEP–WPA3, WPA–WPA2, WPA–WPA3, WPA2–WPA3).

Probability of at least one false positive. If each test uses α = 0.05 and all null hypotheses are true (and the tests are treated as independent), P(at least one false positive) = 1 − (1 − 0.05)^6 = 1 − 0.9500^6 = **0.2649**, about a **26.5%** chance of reporting a difference that does not exist.

Number of pairwise tests	Family-wise error rate
1	0.050
3	0.143
6	0.265
10	0.401

Number of pairwise tests	Family-wise error rate
15	0.537

Why this makes a false discovery likely. Each test carries a 5% chance of a Type I error, and the chances accumulate across tests. Choosing "the pair with the best result" is worse still, because it selects the most extreme of six comparisons, which is exactly the one most likely to be a fluke. Two remedies are used here: an **omnibus one-way ANOVA** that asks one question ("do any means differ?") at $\alpha = 0.05$, followed by **Tukey's HSD**, which controls the family-wise error for all pairwise comparisons. A Bonferroni correction ($\alpha/6 = 0.00833$ per test) would be an alternative.

Task 5.2 – One-Way ANOVA and Post-Hoc Analysis (2 marks)

Vulnerability counts were simulated for 30 access points per encryption type (120 in total; set.seed(5) in the R script, supplied in task5_encryption.csv). True means were 14, 10, 7 and 4 vulnerabilities (WEP, WPA, WPA2, WPA3) with SD 2.5, rounded to whole numbers and floored at 0.

```
print(choose(4, 2)); print(1 - (1 - .05)^6); print(.05 / 6)                # 6 tests,
0.2649, Bonferroni alpha
if (USE_SUPPLIED_DATA) {
  d5 <- rd("task5_encryption.csv"); d5b <- rd("task5_encryption_violated.csv")
} else {
  set.seed(5)
  v <- pmax(round(c(rnorm(30, 14, 2.5), rnorm(30, 10, 2.5), rnorm(30, 7, 2.5),
rnorm(30, 4, 2.5))), 0)
  d5 <- data.frame(enc = rep(c("WEP", "WPA", "WPA2", "WPA3"), each = 30), vulns = v)
  d5b <- d5; d5b$vulns[d5b$enc == "WEP"] <- pmax(round(rnorm(30, 14, 9)), 0)
}
d5$enc <- factor(d5$enc, levels = c("WEP", "WPA", "WPA2", "WPA3")); d5b$enc <-
factor(d5b$enc, levels = levels(d5$enc))
print(aggregate(vulns ~ enc, d5, function(x) c(n = length(x), mean = mean(x), median =
median(x), sd = sd(x), min = min(x), max = max(x))))
m <- aov(vulns ~ enc, d5); print(summary(m))
print(shapiro.test(resid(m))); print(bartlett.test(vulns ~ enc, d5))
ss <- summary(m)[[1]]$"Sum Sq"; print(eta2 <- ss[1] / sum(ss))
print(TukeyHSD(m))
png("plots/t5_diag.png", 1400, 560, res = 150); par(mfrow = c(1, 3))
boxplot(vulns ~ enc, d5, col = "lightblue", ylab = "Vulnerabilities");
qqnorm(resid(m)); qqline(resid(m), col = "red"); plot(fitted(m), resid(m)); abline(h =
0, col = "red"); dev.off()
# 5.3 assumption violation (WEP variance inflated)
```

```
print(tapply(d5b$vulns, d5b$enc, sd))
mb <- aov(vulns ~ enc, d5b); print(summary(mb)); print(shapiro.test(resid(mb)));
print(bartlett.test(vulns ~ enc, d5b))
print(oneway.test(vulns ~ enc, d5b, var.equal = FALSE)); print(kruskal.test(vulns ~
enc, d5b))
png("plots/t5_viol.png", 1200, 500, res = 150); par(mfrow = c(1, 2)); boxplot(vulns ~
enc, d5b, col = "mistyrose"); qqnorm(resid(mb)); qqline(resid(mb), col = "red");
dev.off()
```

Step 1 – Descriptive statistics and boxplots

```
> aggregate(v ~ g, ..., c(n, mean, median, sd, min, max))

      count  mean  median  std  min   max
enc
WEP      30 14.50   15.00 2.67 8.00 18.00
WPA      30 10.20   10.00 2.20 6.00 15.00
WPA2     30  6.60    6.50 2.21 2.00 11.00
WPA3     30  3.43    3.00 2.64 0.00  9.00
```

Screenshot 10: Group-wise descriptive statistics

Encryption	n	Mean	Median	SD	Min	Max
WEP	30	14.50	15.0	2.67	8	18
WPA	30	10.20	10.0	2.20	6	15
WPA2	30	6.60	6.5	2.21	2	11
WPA3	30	3.43	3.0	2.64	0	9

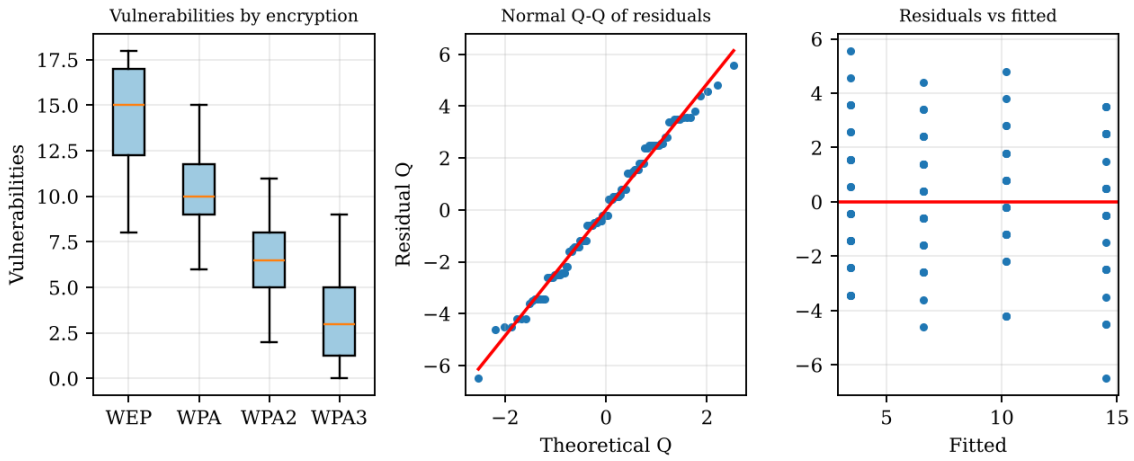


Figure 9: Side-by-side boxplots, normal Q–Q plot of the ANOVA residuals and residuals-versus-fitted plot

Step 2 – Hypotheses

H0: μ_WEP = μ_WPA = μ_WPA2 = μ_WPA3 (all four encryption types have the same mean vulnerability count). H1: at least one mean differs from the others.

Step 3 – Assumption checks

```
> summary(aov(v ~ g)); assumption checks; TukeyHSD(m)

            Df  Sum Sq  Mean Sq  F value   Pr(>F)
enc          3  2041.1   680.37   114.24   0
Residuals  116  690.9    5.96

eta^2 = SSB/SST = 0.747

shapiro.test(resid(m)): W = 0.9887, p-value = 0.4271
bartlett.test(v ~ g):   K-squared = 1.968, df = 3, p-value = 0.5791
leveneTest(v ~ g):      F = 0.788, p-value = 0.503

TukeyHSD(m)   diff      lwr      upr    p adj
WPA-WEP         -4.30    -5.94    -2.66   0.0000
WPA2-WEP        -7.90    -9.54    -6.26   0.0000
WPA3-WEP       -11.07   -12.71    -9.42   0.0000
WPA2-WPA        -3.60    -5.24    -1.96   0.0000
WPA3-WPA        -6.77    -8.41    -5.12   0.0000
WPA3-WPA2       -3.17    -4.81    -1.52   0.0000
```

Screenshot 11: One-way ANOVA table, η², residual checks and TukeyHSD output (original data)

- **Normality of residuals:** Shapiro–Wilk W = 0.9887, p = 0.427, and the Q–Q plot follows the line, so normality is not rejected.

- **Homogeneity of variances:** Bartlett $K^2 = 1.968$, $p = 0.579$; Levene $F = 0.788$, $p = 0.503$. Group SDs are between 2.20 and 2.67, so the equal-variance assumption holds. Independence follows from the design (different access points).

Step 4 – ANOVA table

Source	Df	Sum of Squares	Mean Square	F	p-value
Between (encryption)	3	2041.1	680.37	114.24	< 2.2e-16
Within (residuals)	116	690.9	5.96		
Total	119	2732.0			

Interpretation. SSB (2041.1) is the variation explained by differences between encryption types and SSW (690.9) is the unexplained variation within them. Dividing by their degrees of freedom gives mean squares (680.4 and 5.96), and F = MSB/MSW = **114.2**, with a p-value far below 0.001. We reject H0: at least one encryption type has a different mean number of vulnerabilities. ANOVA does not say which ones differ, so a post-hoc test follows.

Step 5 – Tukey HSD post-hoc comparisons

Comparison	Mean difference	95% CI lower	95% CI upper	Adjusted p
WPA-WEP	-4.30	-5.94	-2.66	2.55e-9
WPA2-WEP	-7.90	-9.54	-6.26	< 2.2e-16
WPA3-WEP	-11.07	-12.71	-9.42	< 2.2e-16
WPA2-WPA	-3.60	-5.24	-1.96	5.20e-7
WPA3-WPA	-6.77	-8.41	-5.12	< 2.2e-16
WPA3-WPA2	-3.17	-4.81	-1.52	1.09e-5

Every pairwise difference is significant after family-wise adjustment and every confidence interval excludes zero. The ordering is WEP > WPA > WPA2 > WPA3 in vulnerabilities, with

each step down significant; WPA3 has the lowest mean and WEP the highest. The smallest gap (WPA3 versus WPA2) is still about 2.8 vulnerabilities per access point.

Step 6 – Effect size (η^2)

η^2 = SSB / SST = 2041.1 / 2732.0 = **0.747**. About 75% of the variation in vulnerability counts is associated with the encryption type. By the usual benchmarks (0.01 small, 0.06 medium, 0.14 large), this is a very large effect: the result is both statistically and practically significant.

Task 5.3 – Assumption Violations and Design Decision (2 marks)

The variance of the WEP group was inflated (data modified: WEP counts redrawn from N(14, 9) with the same mean but a much larger SD; file task5_encryption_violated.csv) while the other three groups were left unchanged.

```
> Task 5.3: variance inflated in WEP

SD by group: WEP=9.34, WPA=2.20, WPA2=2.21, WPA3=2.64
summary(aov): F(3,116) = 32.83, p = 1.89e-15
shapiro.test(resid): W = 0.8364, p = 0.0000
bartlett.test: K-squared = 98.942, p = 2.624e-21   <-- violation detected
leveneTest:    F = 16.666, p = 4.543e-09
oneway.test(var.equal=FALSE): F = 47.255, num df = 3, denom df = 62.28, p = 4.441e-16
kruskal.test: Kruskal-Wallis chi-squared = 67.552, df = 3, p = 1.426e-14
```

Screenshot 12: Assumption checks, ANOVA, Welch ANOVA and Kruskal–Wallis on the modified data

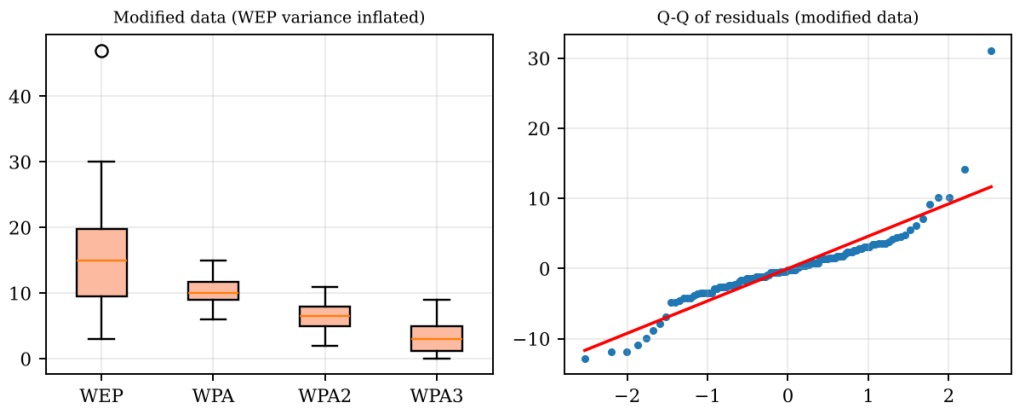


Figure 10: Boxplots and residual Q–Q plot after inflating the WEP variance

Check / test	Original data	Modified data (WEP SD inflated)
SD of WEP / WPA / WPA2 / WPA3	2.7 / 2.2 / 2.2 / 2.6	9.3 / 2.2 / 2.2 / 2.6
Bartlett test	$K^2 = 1.97$, p = 0.579	$K^2 = 98.9$, p = 2.62e-21
Levene test	F = 0.79, p = 0.503	F = 16.67, p = 4.54e-9
Shapiro–Wilk of residuals	W = 0.989, p = 0.427	W = 0.836, p = < 2.2e-16
Classic ANOVA F (df 3, 116)	114.2, p < 2.2e-16	32.8, p 1.89e-15
Welch ANOVA (oneway.test)	not needed	F = 47.3, df = (3, 62.3), p 4.44e-16
Kruskal–Wallis	not needed	$\chi^2 = 67.6$, df = 3, p 1.43e-14
η^2 (classic)	0.747	0.459

- **The violation is detected:** the SD of WEP rises from 2.7 to 9.3 (more than four times the others), and both Bartlett (p 2.62e-21) and Levene (p 4.54e-9) now reject equal variances. The residuals also fail Shapiro–Wilk (p < 2.2e-16). Note that Bartlett is itself sensitive to non-normality, which is why Levene is reported too.
- **Welch's ANOVA and Kruskal–Wallis still reject H0** (Welch F = 47.3, p 4.44e-16; Kruskal–Wallis $\chi^2 = 67.6$, p 1.43e-14). The conclusion that encryption types differ is robust to the violation.
- **Why the classic ANOVA changes:** the classic F falls from 114.2 to 32.8 because the pooled error variance MSW rises from 6.0 to 26.0, dominated by the noisy WEP group. The pooled test also assumes that error variance applies to every group, which is now false, so its p-value and Tukey intervals are unreliable (they are too wide for the quiet groups and too narrow for the noisy one). Welch's test weights each group by the inverse of its own variance and adjusts the denominator degrees of freedom (62.3 instead of 116), so it remains valid. The Kruskal–Wallis test uses ranks and makes no normality assumption, so it also suits the non-normal residuals. After a Welch or

Kruskal–Wallis result, post-hoc comparisons should use Games–Howell or pairwise Wilcoxon tests with a multiplicity correction, instead of Tukey.

Recommendation: which encryption configuration for the whole campus?

Recommendation: WPA3. It has the lowest mean vulnerability count (3.43, against 6.60 for WPA2, 10.20 for WPA and 14.50 for WEP), the differences are significant after correction, the effect is large ($\eta^2 = 0.75$), and the conclusion survives the assumption violation (Welch and Kruskal–Wallis). It also has the strongest security properties: WPA3-Personal replaces the pre-shared-key handshake with Simultaneous Authentication of Equals (SAE), which resists offline dictionary attacks and provides forward secrecy; WPA3 makes Protected Management Frames mandatory (limiting deauthentication attacks); and WPA3-Enterprise offers a 192-bit security mode. WEP is broken by design and should be retired, and WPA (TKIP) is deprecated.

Caveats. (1) The WPA3 group's SD (2.64) is not smaller than the others, so the statistics support "fewer vulnerabilities on average" and not "more consistent". (2) Counts of vulnerabilities do not capture severity. (3) Compatibility: older clients may not support WPA3, and a WPA3/WPA2 transition mode exposes the network to downgrade attacks, so the rollout should proceed by phases, with WPA2-AES (CCMP) as a minimum fallback for legacy devices and early WPA3 firmware patched against the published "Dragonblood" weaknesses. (4) This is a simulated lab data set; the real decision should rest on an assessment of the actual access points.

7. Task 6 – Design Challenge: Complete Security-Data Investigation (13 marks)

Question as set

A security analyst must investigate a real-world security or network dataset (at least 300 observations, at least three numeric variables, at least two categorical variables with one having three or more levels) to answer meaningful questions and support protection decisions, without simply running every function on every variable.

Stages: 1 Understand and clean (2) · 2 Investigation questions (1) · 3 Exploratory data analysis (3) · 4 Predict before testing (3) · 5 Test and validate (2) · 6 Defend the findings (2)

Dataset status (important). The analysis below uses a stand-in network-connection dataset (2,445 records) that has the same structure as NSL-KDD and was generated with set.seed(66) because no internet access was available while this report was prepared. Replace it with the real NSL-KDD file (KDDTrain+.txt, Canadian Institute for Cybersecurity, University of New Brunswick) before submission: set USE_NSLKDD <- TRUE in the script, re-run, and update the numbers. The questions, tests, assumption checks and discussion are written to carry over to the real data; only the values will change.

Stage 1 – Understand and Clean the Data (2 marks)

Dataset name and source. NSL-KDD-style network connection records (each row is one network connection, labelled normal or attack). Real-data source for the final submission: NSL-KDD, Tavallaee et al. (2009), Canadian Institute for Cybersecurity, University of New Brunswick. File of this report: security_dataset_raw.csv.

Variable	Meaning	Type	Role
duration	Connection length in seconds	Continuous (ratio), right-skewed	Response in Q2
src_bytes	Bytes sent from source to destination	Continuous (ratio), heavily skewed	Response in Q1
dst_bytes	Bytes sent from destination to source	Continuous (ratio)	EDA
count	Connections to the same host in the past two seconds	Discrete (count)	EDA
serror_rate	Share of connections with SYN errors (0–1)	Continuous (bounded)	EDA
protocol_type	Transport protocol: tcp, udp, icmp	Nominal, 3 levels	Explanatory in Q2 and Q3
service	Destination service (http, ftp_data, smtp, private, other)	Nominal, 5 levels	EDA

Variable	Meaning	Type	Role
class	Connection label: normal or attack	Nominal, binary	Explanatory in Q1, response in Q3

There are no ordinal variables in this selection. If the real data are used, ordinal candidates are the NSL-KDD difficulty level and the connection flag grouped by severity.

```
> Stage 1: structure and summary of raw data

> str(d)  /  summary(d)
'data.frame': 2445 obs. of 8 variables:
 $ duration      : num  5.0, 3.0, 21.0, 74.0 ...
 $ src_bytes     : num  532.0, 148.0, 36.0, 113.0 ...
 $ dst_bytes     : num  385.0, 112.0, 190.0, 184.0 ...
 $ count         : num  8, 55, 14, 11 ...
 $ serror_rate   : num  0.039, 0.16, 0.08, 0.091 ...
 $ protocol_type : num  udp, icmp, tcp, tcp ...
 $ service       : num  other, private, ftp_data, http ...
 $ class         : num  normal, attack, normal, normal ...

             count   mean     std  min   50%      max
duration    2423.0   19.9    51.3 -1.0   6.0    687.0
src_bytes   2445.0 2684.6 11244.5  1.0 455.0 257491.0
dst_bytes   2407.0 2273.6  9647.6  0.0 309.0 262811.0
count       2445.0   32.3    23.3  4.0  18.0     91.0
serror_rate 2445.0    0.2     0.2  0.0   0.1      0.9
```

Screenshot 13: str() and summary() of the raw data

```
> Stage 1: cleaning log

Rows raw: 2445
Exact duplicates removed: 41  -> 2404
Impossible negative durations set to NA: 6
Missing after fix: duration=28, dst_bytes=38  (<5% each)
Complete-case rows kept: 2340  (removed 64)

Outliers  [IQR raw | IQR on log1p | |z|>3 raw]
duration     269      19      45
src_bytes    316      20      24
dst_bytes    327      11      28
count          0       0       0
```

Screenshot 14: Cleaning log: duplicates, impossible values, missing values and outlier counts

Issue	Found	Decision and justification
Duplicate records	41 exact duplicates	Removed. Exact repeats of the same record add no information and inflate the sample size, which would make tests look more significant than they are
Impossible values	6 negative durations	Set to missing. A negative duration is physically impossible (logging error); they were not “corrected”, because the true value is unknown
Missing values	duration: 28 (22 blank + 6 from the step above); dst_bytes: 38	Less than 2% of rows are affected for each variable, so listwise (complete-case) deletion is used. Imputation would add little and could distort the tails; missingness is assumed to be unrelated to attack status (a limitation)
Outliers (IQR rule, z-score)	Raw: duration 269, src_bytes 316, dst_bytes 327 IQR outliers; after log1p: 19, 20, 11	Kept. Extreme byte counts and durations are genuine traffic (floods, transfers, long sessions) and may be the very attacks of interest. Their influence is reduced by log transformation instead of removal

Rows: 2445 raw → 2404 after removing duplicates → **2340** after complete-case deletion (64 rows lost, 2.7%). The cleaned file is security_dataset_clean.csv. A log(1 + x) transformation was added for the skewed variables (log_src, log_dst, log_dur).

Stage 2 – Formulate Investigation Questions (1 mark)

	Question	Response	Explanatory	Test and why it fits
Q1	Do attack connections transfer a different volume of data from source to destination than normal connections?	log(src_bytes) (continuous)	class (2 groups)	**Independent two-sample t-test (Welch).** One numeric response, one two-level grouping factor, unrelated observations; the log scale makes the byte counts approximately normal
Q2	Does the mean connection duration differ across the three protocols?	log(duration) (continuous)	protocol_type (3 levels)	**One-way ANOVA.** One numeric response and one factor with three levels; testing three means by multiple t-tests would inflate Type I error
Q3	Is the protocol associated with whether a connection is an attack?	class (categorical)	protocol_type (categorical)	**Chi-square test of independence.** Both variables are categorical and the question concerns association between counts

Stage 3 – Exploratory Data Analysis (3 marks)

Univariate analysis

```
> Stage 3: univariate summaries (mean, median, sd, IQR, skewness, kurtosis)

        duration  src_bytes  dst_bytes  count  serror_rate
mean       20.03    2660.36    2276.88  32.41         0.21
median      6.00     451.50     313.00  18.00         0.12
sd         50.70   11250.82    9717.08  23.30         0.21
IQR        15.00    1415.50    1054.25  45.00         0.27
skew        7.17      13.28      13.42   0.50         1.23
kurt       71.07     234.79     269.91   1.49         3.62

After log1p transform:
      log_dur  log_src  log_dst
skew     0.62     0.10     0.11
kurt     3.16     3.03     2.86
```

Screenshot 15: Mean, median, SD, IQR, skewness and kurtosis, before and after the log transformation

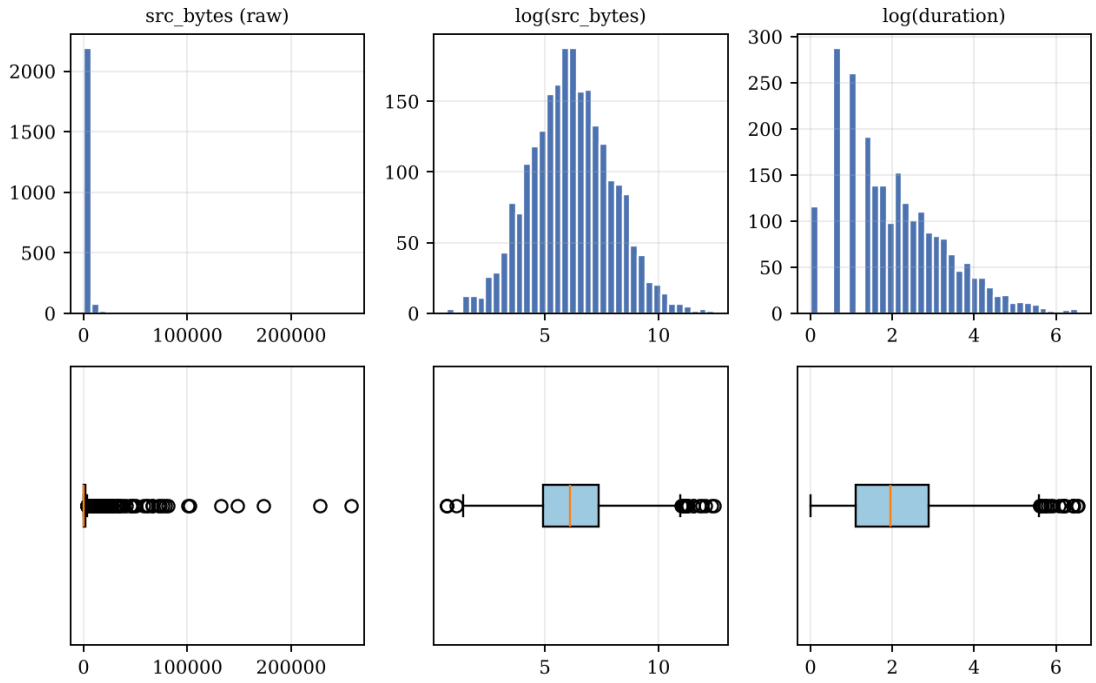


Figure 11: Histograms and boxplots of src_bytes (raw), log(src_bytes) and log(duration)

The raw byte counts are extremely right-skewed (src_bytes: skewness 13.3, kurtosis 235; mean 2660 versus median 452). After log transformation, skewness falls to 0.10 and kurtosis to 3.03, close to the normal values of 0 and 3. Duration behaves similarly (skewness 7.2 raw; 0.62 after log). Only the variables that are relevant to the three questions are explored in detail.

Bivariate analysis

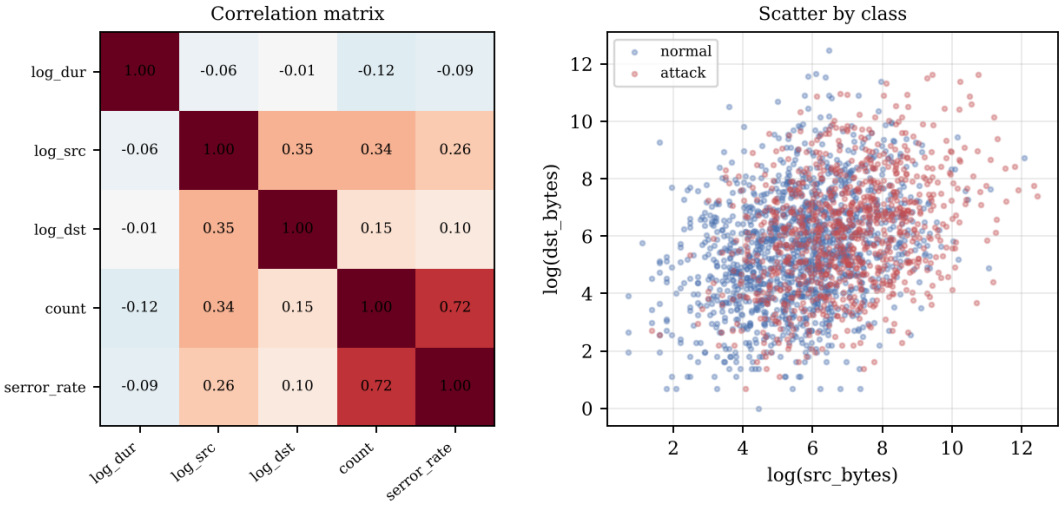


Figure 12: Correlation matrix (left) and scatter plot of log(dst_bytes) against log(src_bytes) coloured by class (right)

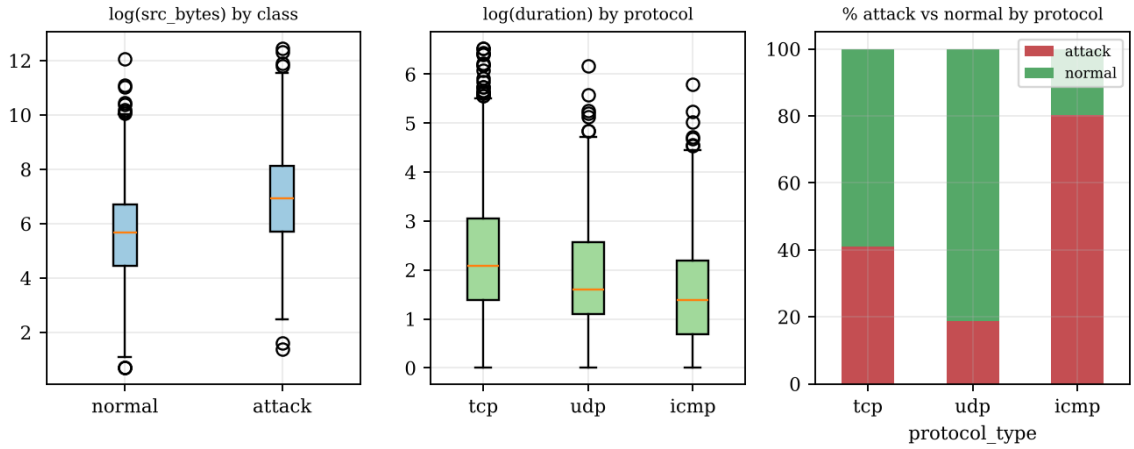


Figure 13: Grouped boxplots by class and protocol, and percentage of attack versus normal by protocol

```
> Stage 3: contingency table and group summaries

Contingency table protocol x class:
class          attack  normal
protocol_type
tcp               641     922
udp               100     435
icmp              194      48

prop.table(, 1):
class          attack  normal
protocol_type
tcp             0.410   0.590
udp             0.187   0.813
icmp            0.802   0.198

log(duration) by protocol:
               count  mean  median   std
protocol_type
tcp             1563 2.266   2.079 1.258
udp              535 1.840   1.609 1.123
icmp             242 1.577   1.386 1.149

Mean by class:
        log_src  count  serror_rate
class
attack    6.945 60.106        0.396
normal    5.623 13.981        0.079
```

Screenshot 16: Contingency table, row proportions and group summaries

Three non-obvious observations

1. **The data need a log scale, and the transformed byte counts separate attack from normal traffic.** Raw bytes are dominated by a few giant transfers (a mean about 6 times the median), which hides the group difference; on the log scale the mean of log(src_bytes) is 6.95 for attacks and 5.62 for normal connections, a gap of about 1.3 log-units, or roughly a 3.8-fold difference in typical volume.
2. **Count and serror_rate move together almost as one signal (r = 0.72), while duration is nearly uncorrelated with the byte counts (r = -0.06).** Connections to the same host in quick succession with SYN errors are the signature of scans and floods. The two variables are largely redundant, so a detection model needs only one of them; duration carries information that the byte counts do not.
3. **Protocol is a strong marker of attack share, and ICMP stands out.** 80% of ICMP connections are attacks, against 41% for TCP and 19% for UDP. ICMP connections are also the shortest (median log-duration 1.39 versus 2.08 for TCP), but the duration

differences are small relative to the within-protocol spread (SD about 1.1–1.3 log-units) and the boxplots overlap heavily.

Stage 4 – Predict Before Testing (3 marks)

The predictions below rely only on the exploratory evidence above, written before any formal test was run.

Investigation	Predicted outcome	EDA evidence	Reasoning
Q1 (t-test)	**Significant, large difference:** attacks send more bytes	Boxplots of log(src_bytes) are clearly shifted; group means differ by about 1.3 log-units, which is roughly 0.8 pooled SDs	Floods and bulk transfers generate far more traffic than interactive sessions, so a large gap in typical volume is plausible
Q2 (ANOVA)	**Significant but small effect:** TCP longer than UDP and ICMP; much overlap	Group medians differ (2.08, 1.61, 1.39 on the log scale) but the boxes overlap heavily and the SDs are similar	With a few thousand records even a modest mean shift is detectable, but protocol will explain only a small part of the variation in duration
Q3 (chi-square)	**Significant, strong association**	Row proportions of attacks: 41% (TCP), 19% (UDP), 80% (ICMP)	Proportions differ by a factor of four across protocols; this cannot plausibly be explained by the sample size

Stage 5 – Test and Validate the Findings (2 marks)

```
# --- Stage 5 tests
# Q1 Welch t-test: log(src_bytes) ~ class
print(shapiro.test(sample(d$log_src[d$class == "attack"], min(500, sum(d$class ==
"attack")))))
print(shapiro.test(sample(d$log_src[d$class == "normal"], 500)))
print(var.test(log_src ~ class, d))
```

```
q1 <- t.test(log_src ~ class, data = d, var.equal = FALSE); print(q1)
sp <- sqrt(((sum(d$class == "attack") - 1) * var(d$log_src[d$class == "attack"]) +
(sum(d$class == "normal") - 1) * var(d$log_src[d$class == "normal"])) / (nrow(d) - 2))
print(c(cohens_d = abs(diff(q1$estimate)) / sp, ratio_geo_means =
exp(abs(diff(q1$estimate)))))
print(wilcox.test(log_src ~ class, d))
# Q2 One-way ANOVA: log(duration) ~ protocol
a <- aov(log_dur ~ protocol_type, d); print(summary(a))
ssq <- summary(a)[[1]]$"Sum Sq"; print(eta2 <- ssq[1] / sum(ssq))
print(shapiro.test(sample(resid(a), 500))); print(bartlett.test(log_dur ~
protocol_type, d))
print(oneway.test(log_dur ~ protocol_type, d, var.equal = FALSE));
print(kruskal.test(log_dur ~ protocol_type, d))
print(TukeyHSD(a)); print(pairwise.t.test(d$log_dur, d$protocol_type, pool.sd = FALSE,
p.adjust.method = "holm"))
# Q3 Chi-square: protocol x class
ch <- chisq.test(ctab); print(ch); print(ch$expected)
print(c(cramers_V = sqrt(unname(ch$statistic) / (sum(ctab) * (min(dim(ctab)) - 1)))))
cat("\nSession info:\n"); print(sessionInfo())
```

```
> Stage 5: t.test / aov+TukeyHSD / chisq.test

Q1 Welch t-test: log(src_bytes) attack vs normal
  t = 18.363, df = 1972.4, p = 1.2e-69
  95% CI of difference: [1.181, 1.463]  (log scale)
  Cohen's d = 0.778; back-transformed ratio of geometric means = 3.75x
  Shapiro (n=500 subsample) p: attack=0.199, normal=0.836; Levene p = 0.327; Wilcoxon p = 9.47e-65

Q2 One-way ANOVA: log(duration) ~ protocol_type
  F(2,2337) = 49.159, p < 2.2e-16, eta^2 = 0.0404
  Shapiro (resid, n=500) p = 1.1e-07; Levene p = 0.00205  (variances unequal -> Welch)
  Welch ANOVA: F(2,612.8) = 52.52, p < 2.2e-16; Kruskal-Wallis chi2 = 99.95, p = 1.98e-22, epsilon^2 = 0.0
  Holm-adjusted Welch pairwise p: udp-tcp 8e-13; icmp-tcp 1.1e-15; icmp-udp 0.0031
  TukeyHSD:
   udp-tcp   diff= -0.427  [ -0.570,  -0.284]  p adj=1.03e-11
   icmp-tcp  diff= -0.689  [ -0.887,  -0.492]  p adj=1.93e-13
   icmp-udp  diff= -0.263  [ -0.484,  -0.042]  p adj=0.0149

Q3 Chi-square: protocol_type x class
  X-squared = 264.64, df = 2, p-value = 3.42e-58
  Cramer's V = 0.336; minimum expected count = 96.7
```

Screenshot 17: Stage 5 test results for Q1 (Welch t-test), Q2 (ANOVA, Welch ANOVA, Kruskal–Wallis, post-hoc) and Q3 (chi-square)

Investigation 1: Welch two-sample t-test

H0: μ(log src_bytes, attack) = μ(log src_bytes, normal). **H1:** the means differ (two-sided, α = 0.05). **Assumptions:** the groups are large (n = 935 and 1405); Shapiro–Wilk on 500-record

subsamples gives p = 0.20 (attack) and 0.84 (normal), so normality holds on the log scale; Levene p = 0.33, so variances are similar (Welch is used regardless). **Result:** t(1972.4) = 18.36, p = 1.2e-69; difference in means = 1.32 log-units, 95% CI [1.18, 1.46]; Cohen's d = 0.78 (medium-to-large). Back-transformed, the geometric mean of src_bytes is 3.75 times higher for attacks. The Wilcoxon rank-sum test agrees (p = 9.5e-65). **Reject H0.**

Investigation 2: One-way ANOVA with post-hoc tests

H0: μ_tcp = μ_udp = μ_icmp (log duration). **H1:** at least one mean differs. **Assumptions:** residual normality is rejected in the formal test (Shapiro–Wilk on 500 residuals, p = 0.0e+0), because durations are whole seconds with many small values, and Levene's test finds unequal variances (F = 6.21, p = 0.0021). With these violations the classical F-test is not the safest choice, so **Welch's ANOVA** and **Kruskal–Wallis** are used as the primary tests, with the classical ANOVA reported for comparison. **Classical ANOVA:** F(2, 2337) = 49.2, p < 2.2e-16, η^2 = 0.040. **Welch ANOVA:** F(2, 612.8) = 52.5, p < 2.2e-16. **Kruskal–Wallis:** $\chi^2(2)$ = 99.9, p = 2.0e-22, ε^2 = 0.042. All three tests agree. **Post-hoc (Tukey HSD; Holm-adjusted Welch pairwise tests agree):**

Pair	Difference in mean log-duration	95% CI	Adjusted p
udp-tcp	-0.427	[-0.570, -0.284]	1.0e-11
icmp-tcp	-0.689	[-0.887, -0.492]	1.9e-13
icmp-udp	-0.263	[-0.484, -0.042]	1.5e-2

All three pairs differ: TCP connections last longest, then UDP, then ICMP (the largest gap, TCP versus ICMP, is 0.69 log-units, about 50% shorter). **Effect size:** η^2 = 0.040 means that protocol explains only about 4% of the variation in log-duration, a small-to-medium effect.

Investigation 3: Chi-square test of independence

H0: protocol_type and class are independent. **H1:** they are associated. **Assumptions:** observations independent; every expected cell count is at least 97 (rule of thumb: ≥ 5), so the chi-square approximation is valid. **Result:** $\chi^2(2)$ = 264.6, p = 3.4e-58; Cramér's V = 0.34 (a moderate-to-strong association for a 2 × 3 table). **Reject H0.** Row proportions: attack share 41% (TCP), 19% (UDP), 80% (ICMP).

Comparison with the Stage 4 predictions

Investigation	Predicted	Actual	Verdict
Q1 t-test	Significant, large	Significant; d = 0.78	Supported (the effect is medium-to-large)
Q2 ANOVA	Significant but small	Significant; $\eta^2 = 0.040$	Supported; the assumption violations did not change the conclusion
Q3 chi-square	Significant, strong	Significant; V = 0.34	Supported

Stage 6 – Defend the Findings (2 marks)

Were the predictions supported, and why? All three were supported, because they were grounded in specific EDA evidence: separated boxplots for Q1, heavily overlapping boxes for Q2 (predicting a small effect rather than only "significant"), and very different row proportions for Q3. The most informative prediction was Q2, because it anticipated the difference between detecting an effect and the size of that effect.

Statistical versus practical significance. With about 2,300 records almost any real difference gives $p \approx 0$. P-values show that an effect exists, not that it is large. Q1: attacks send about 3.8 times more bytes, but d = 0.78 means that the distributions overlap considerably (using the normal approximation, a random attack connection has more bytes than a random normal one only about 71% of the time), so byte volume is a useful feature but not a sufficient detector on its own. Q2: statistical significance is overwhelming, but $\eta^2 = 0.040$ means protocol explains only 4% of the variation in duration, so a duration alert threshold would not need to be protocol-specific. Q3: V = 0.34 and attack shares from 19% to 80% are operationally meaningful.

Dataset limitations.

- **Sampling bias and base rates:** the attack share (about 40%) is far higher than in real traffic, where attacks are rare; the proportions therefore cannot be used as probabilities of attack in a live network, and a classifier trained on such data would give many false alarms in practice. NSL-KDD is also a laboratory data set based on 1999 attack types.

- **Missing variables:** no timestamps, source and destination addresses, payload, user or asset criticality are available, so attack severity, campaign structure and business impact cannot be assessed.
- **Potential confounders:** protocol is confounded with service (for example ICMP traffic is concentrated in a few services), attack type (flooding versus probing) and time. The Q1 and Q2 differences may partly reflect attack type and not class or protocol as such.
- **Cleaning choices:** complete-case deletion assumes that the data are missing at random with respect to class, which cannot be verified. Log transformation changes the question to one about geometric means (ratios).
- **Multiple tests and independence:** connections from the same attacker or session are not independent, which makes p-values too small.

What a decision-maker can reasonably conclude. (1) Protocol type is a strong, actionable marker of attack likelihood in this data, and ICMP traffic deserves priority monitoring and rate-limiting. (2) Byte volume per connection is higher for attacks and is a useful input to detection rules but should be combined with other features (such as count and serror_rate). (3) Connection duration differs somewhat between protocols but is a weak discriminator.

What cannot be concluded. That protocol or traffic volume **causes** attacks; that these proportions hold in another network; that byte counts alone can separate attacks from normal traffic; or that the ranking of protocols would stay the same under another attack mix.

Protection recommendations. Prioritise monitoring and rate-limiting of the high-risk protocol (ICMP) and the corresponding firewall rules; add a volume-based alert on log(src_bytes) together with the count and serror_rate features; validate any rule on a realistic, imbalanced data set before enabling blocking; and treat the findings as hypotheses to test on current traffic.

8. Overall Conclusions, Protection Recommendations and Viva Preparation

8.1 Summary of Findings and Recommendations

Task	Key finding	Protective recommendation
1	The mean overstates typical response time on skewed data; the median and the 95th percentile together describe the user experience	Publish median and P95 on the status page; investigate slow-tail causes (resource exhaustion, abusive clients)
2	Equal averages hide very different stability: Config A is most consistent and compliant; B is unstable; C has dropouts	Deploy Config A; monitor the percentage of samples outside ±5% and CV; investigate C's dropouts (interference, roaming, deauthentication)
3	Patch age and open ports are positively related to vulnerabilities; hardening is negatively related (weaker than predicted); failed logins unrelated; Windows hosts and hosts without a firewall have more findings	Prioritise patching of the oldest hosts, reduce open ports, enforce firewalls and baseline hardening; use failed logins for attack detection, not for vulnerability ranking
4	Study 1 shows no evidence of a difference from the 300 ms target; the new WAF rule set blocks more; the hardening baseline reduces vulnerabilities	Adopt the new WAF rules after checking false positives; roll out the hardening baseline and keep a control group when re-measuring
5	The multiple-testing risk is about 26% for six uncorrected tests; encryption types differ greatly ($\eta^2 \approx 0.75$); conclusions are robust to unequal variances	Standardise on WPA3 (with a controlled transition for legacy devices); retire WEP and WPA

Task	Key finding	Protective recommendation
6	Attack traffic has far larger byte volumes; protocol is strongly associated with attack share; duration differences are small	Prioritise monitoring of the highest-risk protocol; combine volume with count and serror_rate; validate on real, imbalanced traffic

8.2 Viva Preparation: Questions You Should Be Able to Answer

Likely question	Model answer
Why did you choose the median over the mean for the response-time figure?	The data are right-skewed, so a few slow requests inflate the mean. The median is robust (50% breakdown point) and easy to explain; I pair it with the 95th percentile to show the tail.
When do SD and IQR give different answers?	When the data contain outliers or are skewed. SD squares deviations and is pulled by dropouts (Config C: SD > IQR by a large factor); the IQR describes only the middle 50%.
Why use the coefficient of variation in Task 2.5?	The targets differ (100 versus 10 Mbps), so raw SD is not comparable. CV = SD/mean puts every configuration on a relative scale.
Pearson or Spearman?	Pearson measures linear association and is sensitive to outliers; Spearman measures monotonic association on ranks and is robust. Where they differ (patch_age, outliers in failed_logins), I prefer Spearman.
How do you choose between the three t-tests?	By the design: one group against a constant (one-sample), two unrelated groups (independent/Welch), the same units measured twice (paired).
Why a one-sided test in Study 3 but two-sided in Studies 1 and 2?	The Study 3 question is directional (“reduce”); the others ask whether the mean is “different”. I decided the direction from the question before seeing the data.

Likely question	Model answer
Why not run six t-tests for four encryption types?	The family-wise error rate rises to 1 − 0.95^6 ≈ 26.5%. I use ANOVA first and Tukey HSD afterwards.
What did you do when ANOVA assumptions failed?	I detected the problem with Levene/Bartlett and Shapiro–Wilk, and switched to Welch's ANOVA and Kruskal–Wallis, which still rejected H0.
What is the difference between a significant and an important result?	Significance depends on sample size; effect sizes (d, η^2, V) measure importance. In Task 6, Q2 is hugely significant but explains about 4% of the variation.
What are the limitations of your Task 6 conclusions?	Sampling bias and unrealistic attack base rates, missing variables and confounders (service, attack type, time), and dependent observations; the findings are associations, not causes.

Appendix A – Reproducibility, File List and Software

Item	Details
R script	ITA14_Assignment4A_analysis.R: one script, runs top to bottom from the working directory containing the data/ folder
Random seeds	Task 1: 14; Task 2: 24 and 25 (IoT); Task 3: 3; Task 4: 4; Task 5: 5 (and a modified WEP group); Task 6: 66. The datasets in data/ are the fixed values used for every number in this report; running the script with USE_SUPPLIED_DATA <- FALSE regenerates them in R with the stated seeds
Raw datasets	task1_response_times.csv, task2_throughput.csv, task2_c_iot.csv, task3_hosts.csv, task4_study1.csv, task4_study2.csv, task4_study3.csv, task5_encryption.csv, task5_encryption_violated.csv, security_dataset_raw.csv
Cleaned dataset	security_dataset_clean.csv (Task 6, after removing duplicates, impossible values and incomplete rows)

Item	Details
Plots	plots/ folder, written by the script (histograms, boxplots, scatter plots, heatmap, bar charts, Q–Q plots)
Packages	Base R and moments (skewness, kurtosis). No other package is required
Significance level	$\alpha = 0.05$ throughout; assumption checks and effect sizes are reported for every test

Appendix B – References

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